

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Akuakultur Department of Aquaculture

AKU3111	Iktiologi <i>Ichthyology</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi topik kepelbagaian biologi dan ekologi ikan. Pengkelasan kumpulan ikan dengan ciri primitif dan moden diperincikan secara umum melalui perbandingan anatomi dan struktur morfologi. Tabiat dan keupayaan penyesuaian ikan di ruang geografi yang berbeza turut dibincangkan. <i>This course covers topics on the diversity of fish biology and ecology. The classification of fish groups with primitive and modern characteristics is generally detailed through a comparison of anatomy and morphological structure. Behaviors and adaptive ability of fishes in diverse geographical spaces are also discussed.</i>		
AKU3121	Biologi Invertebrat Akuatik <i>Biology of Aquatic Invertebrates</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi aspek diversiti, ekologi, ciri biologi dan kepentingan komersial invertebrat akuatik serta hubungannya dengan sektor akuakultur. Pertalian antara keperluan fisiologi, adaptasi perlakuan dan kemandirian invertebrat dengan bentuk struktur dan sistem biologinya ditekankan. <i>The course covers aspects on diversity, ecology, biological characteristics and commercial importance of aquatic invertebrates in relation to the aquaculture sector. Relationship between physiological needs, behavioral adaptation and survival of invertebrates with its structural forms and biological system is emphasised.</i>		
AKU3141	Botani Akuatik <i>Aquatic Botany</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi taksonomi dan pengkelasan, ciri morfologi penting bagi setiap kumpulan utama tumbuhan akuatik berdasarkan persekitaran air tawar dan marin. Kegunaan, kepentingan dan ancaman terhadap tumbuhan akuatik dibincangkan. <i>This course covers the taxonomy and classification, important characteristic for each main group of aquatic plants based on the freshwater and marine environments. Utilization, importance and threats to aquatic plants are discussed.</i>		
AKU3200	Prospek Akuakultur <i>Aquaculture Prospects</i>	1 (1+0)
Prasyarat : Tiada		
Kursus ini memberikan pendedahan mengenai konsep dan prospek industri akuakultur dalam konteks keterjaminan makanan. Penilaian amalan akuakultur yang bertanggung jawab secara sistematik diperingkat industri dan komuniti turut diberikan penekanan. <i>This course provides exposure to the concepts and prospects of the aquaculture industry in the context of food security. The evaluation of responsible aquaculture practices in a systematic way at the industry and community levels are also emphasized.</i>		

AKU3804	Statistik dan Penyelidikan Dalam Akuakultur <i>Statistics and Research in Aquaculture</i>	4 (3+1)
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Prasyarat : Tiada

Kursus ini merangkumi asas statistik dan analisis data dalam akuakultur dan perikanan. Kursus ini menekankan interpretasi hasil analisis data dalam situasi sebenar, selain menekankan rekabentuk eksperimen dengan menggunakan formulasi statistik yang sesuai. Penilaian impak dan risiko jangka panjang untuk kelestarian sektor akuakultur dan perikanan juga diberi penekanan.

This course covers the fundamental of statistics and data analysis in aquaculture and fisheries. This course emphasises on interpretation of results of data analysis in the real-life situation, besides emphasises on the experimental design using appropriate statistical formulation. Long-term impact and risk assessment for the sustainability of the aquaculture and fisheries sectors is also emphasized.

AKU3803	Fisiologi Ikan dan Biokimia <i>Fish Physiology and Biochemistry</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi sistem, fungsi dan komponen utama fisiologi untuk homeostasis berterusan dalam hidupan akuatik. Proses biokimia dan fisiologi ikan yang relevan dengan persekitaran dan pelbagai laluan metabolisme dalam organisma akuatik ditekankan.

This course covers the main system, function and physiological components for continuous homeostasis in aquatic life. Biochemical and fish physiological processes relevant to the environment and various metabolic pathways in aquatic organisms are emphasized.

AKU3202	Prinsip Akuakultur <i>Principles Of Aquaculture</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi prinsip asas akuakultur untuk keterjaminan makanan. Proses yang terlibat dalam pra-produksi, produksi, dan pasca-produksi akuakultur yang lestari mengikut piawaian yang ditetapkan dibincangkan.

This course covers basic principles of aquaculture for food security. The processes involved in pre-production, production and post-production of sustainable aquaculture according to standard practices are discussed.

AKU3211	Kultur Ikan <i>Finfish Culture</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi kultur dan pengurusan ikan makanan, terutamanya spesies air tawar, dengan penekanan kepada amalan mampan bagi memastikan keterjaminan makanan. Penjagaan dan pengurusan ikan secara sistematik dan melalui kolaborasi dengan komuniti turut diberi perhatian

This course covers the culture and management of food fish, especially freshwater species, with an emphasis on sustainable practices to ensure food security. Systematic care and management of fish and through collaboration with the community are also given attention.

AKU3311	Genetik dan Pembiakbakaan Ikan <i>Fish Genetics And Breeding</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi konsep genetik, pembiakbakaan ikan, penghibridan dan manipulasi genetik. Penyelesaian masalah dalam pemilihan baka ikan secara bersepadu dan etika bagi kelestarian akuakultur dibincangkan.

This course covers concepts of fish genetics, breeding, hybridisation and genetic manipulation. Problems solving for selective breeding in fish through integrated and ethical approaches for sustainability of aquaculture are discussed.

AKU3312	Teknologi Pembenihan Ikan dan Pengurusan Fri <i>Fish Seed Technology And Fry Management</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi biologi pembiakan ikan, teknik pembenihan ikan, larvikultur, pengurusan, pembungkusan dan pengangkutan fri. Permasalahan dalam pembiakan ikan dibincangkan.		
<i>This course covers fish reproductive biology, fish seed production techniques, larviculture, management, packaging and transportation of fry. Issues in fish breeding are discussed.</i>		
AKU3403	Pemakanan Akuakultur <i>Aquaculture Nutrition</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi amalan penyediaan dan pemberian makanan dalam akuakultur. Nutrisi dalam bahan makanan untuk pertumbuhan optimum haiwan akuatik, isu semasa dan kekangan dalam pemakanan akuakultur diberi penekanan.		
<i>This course covers preparation and feeding practices in aquaculture. Nutrition in feed ingredients for optimal growth of aquatic animals, current issues and limitations in aquaculture nutrition are emphasized.</i>		
AKU3502	Pengurusan Kesihatan Ikan <i>Fish Health Management</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi pengenalan dan konsep patologi dan mikrobiologi berhubung dengan penyakit ikan, tatacara diagnosis, kawalan dan rawatan. Kepentingan amalan pengurusan yang baik, pengawalan dan pencegahan penyakit dalam akuakultur dibincangkan.		
<i>This course covers introduction and concepts of pathology and microbiology in relation to fish diseases, diagnostic procedures, control and treatment. The importance of good husbandry, control and prevention of diseases in aquaculture are discussed.</i>		
AKU3603	Kualiti Air dan Tanah dalam Akuakultur <i>Water and Soil Quality in Aquaculture</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi teori dan penentuan kualiti air dan tanah untuk pemilihan tapak. Analisis dan tafsiran kimia dan fizikal air dan tanah diberi penekanan.		
<i>This course covers the theory and determination of water and soil quality for site selection. Analyses and interpretation of the physical, and chemicals of water and soil are emphasized.</i>		
AKU3604	Ekologi Akuatik <i>Aquatic Ecology</i>	3(2+1)
Prasyarat : Tiada		
Kursus ini merangkumi konsep ekologi akuatik dan teknik ukuran parameter ekologi. Impak terhadap ekosistem akuatik dan kaedah pemantauan serta pemulihan dibincangkan.		
<i>This course covers concepts in aquatic ecology and ecological parameters measurement techniques. Impacts towards aquatic ecosystem as well as the monitoring and the remediation techniques are discussed.</i>		
AKU3805	Pasca Tuai Dalam Akuakultur <i>Postharvest In Aquaculture</i>	2 (1+1)
Prasyarat : Tiada		
Kursus ini merangkumi pengendalian pelbagai produk akuakultur, pemprosesan dan pengawalan kualiti. Produk akuatik ditambah nilai dan sampingan dibincangkan.		
<i>This course covers handling of various types of aquatic produce, processing and quality control. Value added product and byproduct of aquatic produce are discussed.</i>		

AKU3806	Inovasi dalam Akuakultur <i>Innovation in Aquaculture</i>	3 (1+2)
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Prasyarat : Tiada

Kursus ini merangkumi teori dan inovasi produk, kekangan dan cabaran dalam sektor akuakultur. Kreativiti dalam pembangunan penyelesaian inovatif melalui pendekatan pemikiran kritis terhadap isu-isu semasa yang berkaitan dengan kelestarian industri akuakultur ditekankan.

This course covers theory and product innovation, constraints and challenges in the aquaculture sector. Creativity in developing innovative solutions through a critical thinking approach to current issues related to the sustainability of aquaculture industry are emphasised.

AKU4104	Penilaian dan Pengurusan Sumber Perikanan <i>Fisheries Resource Assessment and Management</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini merangkumi pengelasan sumber perikanan, teknik penangkapan ikan dan polisi, dinamik populasi dan pengurusan perikanan. Aplikasi perisian komputer dalam penilaian stok ikan dan pengurusan sumber perikanan yang lestari dibincangkan.

This course covers the classification of fisheries resource, fishing technique and policy, population dynamic and management of fisheries. Application of computer software in fish stock assessment and sustainable fisheries resource management are discussed.

AKU4105	Limnologi <i>Limnology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi komponen biotik dan produktiviti biologi dalam ekosistem air tawar serta proses fiziko-kimia yang mempengaruhinya. Kepentingan dan aplikasi limnologi dalam pengurusan perikanan darat dan akuakultur ditekankan.

This course covers the biotic components and biological productivity of freshwater ecosystems and the physico-chemical processes affecting them. The importance and application of limnology in inland fisheries and aquaculture management are emphasized.

AKU4106	Oseanografi <i>Oceanography</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi faktor fizikal, kimia dan biologi dalam ekosistem lautan. Impak aktiviti antropogenik, perubahan iklim dan penyelesaian kepada kesan tersebut dibincangkan untuk kelestarian perikanan dan akuakultur.

This course covers physical, chemical and biological factors in marine ecosystems. Impact of anthropogenic activity, climate change and solutions to effect are discussed for the sustainability of fisheries and aquaculture.

AKU4205	Kultur Ikan dan Tumbuhan Hiasan Akuatik <i>Culture of Ornamental Fish and Aquatic Plants</i>	3 (1+2)
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Prasyarat : Tiada

Kursus ini merangkumi pengkulturan ikan dan tumbuhan hiasan akuatik terpilih. Kaedah dan manipulasi dalam pengkulturan serta pengurusan ikan dan tumbuhan hiasan akuatik yang berpotensi untuk pengkomersialan dibincangkan.

This course covers the cultures of selected ornamental fishes and aquatic plants. Methods and manipulation of cultures as well as management of potential ornamental fishes and plants for commercialization are discussed.

AKU4206	Sistem Pengeluaran Akuakultur <i>Aquaculture Production Systems</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi reka bentuk, pemilihan peralatan asas dan pembinaan pelbagai sistem akuakultur. Pengurusan dan kecekapan sistem untuk operasi optimum dibincangkan. <i>This course covers design, basic equipment selection and construction of various aquaculture systems. The management and efficiency of the systems for optimal operation are discussed.</i>		
AKU4207	Kultur Spesies Unik <i>Unique Species Culture</i>	2 (2+0)
Prasyarat : Tiada		
Kursus ini merangkumi teknik kultur dan pengurusan pelbagai spesies akuatik unik di perairan Malaysia. Spesies unik marin dan air tawar yang belum dikultur secara komersial dalam industri akuakultur di Malaysia ditekankan. Kegunaan produk dari setiap spesies unik dibincangkan. <i>The course encompasses the culture techniques and management of various unique aquatic species in Malaysian waters. Marine and freshwater unique species which are yet to be cultured commercially in Malaysian aquaculture industry are emphasised. Usage of products from each unique species are discussed.</i>		
AKU4222	Kultur Krustasea <i>Crustacean Culture</i>	3 (1+2)
Prasyarat : Tiada		
Kursus ini merangkumi biologi dan kemahiran pengkulturan krustasea komersial terpilih. Isu dan permasalahan dalam industri kultur haiwan krustasea dibincangkan. <i>This course covers the biology and skills of culture on selected commercial crustaceans. Issues and problems in crustacean culture industry are discussed.</i>		
AKU4232	Kultur Bivalvia <i>Bivalve Culture</i>	3 (1+2)
Prasyarat : Tiada		
Kursus ini menekankan aspek pengkulturan pelbagai spesies bivalvia komersial dengan menilai sistem, prosedur dan pengurusan. Prosedur pengkulturan dan pengurusan ternakan juga diterangkan, bersama prinsip biologi, ekologi, dan etika akuakultur mampan untuk spesies bivalvia. <i>This course emphasizes aspects of the culture of various commercial bivalve species by evaluating systems, procedures and management. Culture and stock management procedures are also described, along with the biological, ecological, and ethical principles of sustainable aquaculture for bivalve species.</i>		
AKU4242	Kultur Flora Akuatik <i>Culture of Aquatic Flora</i>	3 (1+2)
Prasyarat : Tiada		
Kursus ini merangkumi pelbagai teknik pembiakan dan pengkulturan flora akuatik air masin dan air tawar. Pengurusan pelbagai pengkulturan flora akuatik dan faktor yang mempengaruhi pertumbuhan diberi penekanan. Kepentingan flora akuatik komersial terpilih dari segi ekonomi dan ekologi juga dibincangkan. <i>This course covers various propagation and culture techniques of aquatic flora. Management various cultures of aquatic flora and factors affecting growth in cultural environments are emphasized. The economic and ecological importance of selected commercial aquatic flora is also discussed.</i>		
AKU4403	Kultur Makanan Hidup <i>Live Food Culture</i>	3 (1+2)
Prasyarat : Tiada		
Kursus ini merangkumi pelbagai teknik inovatif dalam pengeluaran makanan hidup yang utama untuk akuakultur. Pengurusan strategik penyediaan makanan hidup dan faktor yang mempengaruhi pertumbuhan makanan hidup dalam hatcheri diberi penekanan.		

This course covers the innovative techniques in the production of key live food for aquaculture. The strategic management of live food preparation and the factors affecting the growth of live food in hatchery are emphasized.

AKU4302	Endokrinologi Ikan <i>Fish Endocrinology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi biosintesis kimia, struktur dan fungsi hormon daripada kelenjar endokrin. Permasalahan berkaitan dengan endokrinologi ikan ditekankan. Tindakbalas dan peranan hormon terhadap tekanan fizikal dan kimia dalam persekitaran akuatik dibincangkan.

This course covers the chemical biosynthesis, structures and functions of hormones from the endocrine glands. Problems related to fish endocrinology are emphasised. Reaction and roles of hormones on physical and chemical stresses in aquatic environments are discussed.

AKU4404	Teknologi Makanan Akuakultur <i>Aquafeed Technology</i>	3 (1+2)
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Prasyarat : Tiada

Kursus ini meliputi pelbagai rumusan makanan ikan dan teknologi pembuatan. Setiap peringkat pemprosesan dan pengawalan mutu makanan untuk industri akuakultur turut ditekankan.

This course covers various fish food formulations and manufacturing technology. Every stage of feed processing and quality control for the aquaculture industry are emphasized.

AKU4503	Mikrobiologi Akuatik <i>Aquatic Microbiology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi peranan, kesan, teknik pengesanan dan hubungan mikroorganisma dalam akuakultur. Kepentingan, aplikasi dan isu semasa mikrobiologi dalam akuakultur dibincangkan.

This course covers the roles, effects, identification techniques and relations of microorganisms in aquaculture. The importance, applications and current issues of microbiology in aquaculture are discussed.

AKU4504	Imunologi Ikan <i>Fish Immunology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi konsep, jenis, tindak balas dan pengawal ubah imun dalam sistem imuniti ikan. Kepentingan dan aplikasi imunologi ikan untuk pencegahan penyakit dalam akuakultur dibincangkan.

This course covers the concepts, types, responses and immunomodulators in fish immune system. The importance and application of fish immunology for diseases prevention in aquaculture are discussed.

AKU4602	Ekologi dan Pengurusan Tanah Basah <i>Ecology and Management of Wetlands</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi pelbagai ekosistem tanah basah semulajadi dan buatan, fungsi dan kepentingannya. Pengurusan pelbagai ekosistem tanah basah dan faktor yang mempengaruhi sumber hidupnya akibat perubahan iklim dan persekitaran dibincangkan.

This course covers the various natural and constructed wetland ecosystems, function and their importance. Management of various wetland ecosystems and factors affecting their living resources due to climate change and environment are discussed.

AKU4703	Teknologi Maklumat dan Komunikasi Dalam Akuakultur <i>Information and Communications Technology in Aquaculture</i>	3 (1+2)
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Prasyarat : Tiada

Kursus ini merangkumi kegunaan dan kepentingan teknologi maklumat dan komunikasi dalam akuakultur. Kesesuaian teknologi maklumat dan komunikasi dalam operasi dan pengurusan akuakultur dibincangkan.

This course covers the utilization and importance of information and communications technology in aquaculture. The suitability of communications and information technology in aquaculture operation and management is discussed.

AKU4702	Pengeluaran Akuakultur <i>Aquaculture Production</i>	4 (3+1)
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Prasyarat : Tiada

Kursus ini merangkumi penjagaan dan pengurusan akuakultur dari pembenihan hingga pasca tuai. Penggunaan teknologi pintar dalam pengeluaran ternakan dan pasca tuai diberi penekanan. Prosedur amalan akuakultur baik dalam penjagaan dan pengurusan ternakan, pengendalian pra tuai dan pasca tuai serta pengurusan sisa turut dibincangkan.

This course covers the operation and management of aquaculture productions from nursery to post-harvest. The use of smart technology in aquaculture production and post-harvest is emphasized. Good aquaculture practices in handling and managing fish, pre-harvest, post-harvest handling, and waste management are also discussed.

AKU4805	Perikanan Rekreasi Dalam Ekopelancongan Perairan Darat <i>Recreational Fisheries in Inland Waters Ecotourism</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi konsep biologi dan sosio-ekonomi pengurusan perikanan rekreasi sebagai alternatif dalam pengurusan sumber. Pelaksanaan dan operasi perikanan rekreasi di Malaysia dan aspek eko-pelancongan dibincangkan.

This course covers on the concept of biology and socio-economic in recreational fisheries management as alternatives for resource management. Implementation and operation of recreational fisheries in Malaysia and aspect of eco-tourism are discussed.

AKU4807	Polisi dan Akta Untuk Perikanan dan Akuakultur <i>Policy and Acts for Fisheries and Aquaculture</i>	2(2+0)
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Prasyarat : Tiada

Kursus ini merangkumi polisi, akta dan perundangan yang berkaitan dengan industri perikanan dan akuakultur di Malaysia. Kesesuaian pelaksanaan melalui penilaian terhadap peraturan dan undang-undang dalam sektor perikanan turut diberi penekanan.

This course covers policies, acts and legislation related to the fisheries and aquaculture industry in Malaysia. The appropriateness of implementation through the assessment of regulations and laws in the fisheries sector is also emphasized.

AKU4812	Teknologi Hasil Akuatik <i>Aquatic Product Technology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi pengendalian pasca tuai, teknologi pemprosesan, serta ciri fisiko-kimia dan mikrobiologi dalam produk hasil. Aspek pengawetan, pembungkusan, penyimpanan, dan pengangkutan untuk menjamin nilai komersial serta kualiti produk dari segi jangka hayat, pengesanan, dan biosekuriti makanan juga diberi penekanan.

This course encompasses postharvest handling, processing technology, and physicochemical and microbiological properties of products. Aspects on preservation, packaging, storage, and transportation to ensure the commercial value and quality of products in terms of shelf life, traceability, and food biosecurity are also emphasised.

AKU4902	Latihan Industri <i>Industrial Training</i>	12 (0+12)
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Prasyarat : Persetujuan Penyelaras

Kursus ini melibatkan penempatan pelajar di ladang akuakultur swasta atau kerajaan. Pendedahan praktikal kepada keadaan pekerjaan serta amalan pengurusan sebenar dalam bidang akuakultur dan perikanan diberikan penekanan melalui tugas berdasarkan pembangunan mampan sektor akuakultur.

This course involves student attachment at a private or government aquaculture farm. A practical exposure to the actual working conditions and management practices carried out in the field of aquaculture and fisheries are emphasised through assignments on sustainable development of aquaculture sector.

AKU4903	Disertasi Bachelar <i>Bachelor Dissertation</i>	6 (0+6)
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Prasyarat : Tiada

Kursus ini merangkumi sorotan bahan rujukan, penggunaan teknik penyelidikan yang sesuai, pengumpulan dan analisis data, penafsiran keputusan, perbincangan, kesimpulan dan keaslian kajian saintifik dalam projek penyelidikan. Pembinaan kesedaran sendiri dalam mengenal pasti nilai, kepercayaan dan tanggungjawab sebagai penyelidik dalam bidang akuakultur juga diberi penekanan.

This course covers literature review, use of appropriate research techniques, data collection and analyses, interpretation of results, discussion, conclusion and originality of scientific studies in a research project. Emphasis is also placed on building self-awareness in identifying values, beliefs and responsibilities as researchers in the field of aquaculture.

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Sains Haiwan
Department of Animal Science

SHW3002	Pertanian Ternakan / <i>Animal Agriculture</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi kepentingan pertanian ternakan kepada tamadun manusia, di samping memperkenalkan elemen sistem pertanian masa kini dan konsep pertanian lestari. Aspek asas dalam kebajikan dan kesihatan haiwan, serta pengenalan kepada amalan pengurusan haiwan yang baik juga akan dibincangkan. <i>This course encompasses the importance of animal agriculture to human civilization, apart from introducing students to the elements of current agricultural practices and sustainable agriculture. Basics animal welfare and animal health, as well as good animal health practices are also discussed</i>		
SHW3104	Pengurusan Kesihatan Haiwan / <i>Animal Health Management</i>	3 (2+1)
Prasyarat : SHW3002 Kursus ini merangkumi konsep asas kesihatan dan penyebab serta kesan penyakit terhadap haiwanmakanan. Elemen fisiologi, epidemiologi, mikrobiologi, parasitologi, pemakanan dan pengurusan pengeluaran diintegrasikan dalam pendekatan pengurusan kesihatan haiwan. Amalan perladangan baik, penyakit pekerjaan berkaitan industri haiwan, dan prinsip zoonosis turut dibincangkan. <i>This course encompasses basic concepts of health and causes including effects of diseases on food animals. Elements of physiology, epidemiology, microbiology, parasitology, nutrition, and production management are integrated with the approach of animal health management. Good farming practices, occupational diseases related to the animal industry, and principles of zoonosis are discussed.</i>		
SHW3105	Kelakuan dan Kebajikan Haiwan / <i>Animal Behaviour and Welfare</i>	3(3+0)
Prasyarat : SHW3200 Kursus ini meliputi pelbagai aspek kelakuan haiwan, penyebab kelakuan dan implikasinya ke atas pengurusan, produksi, kesihatan dan kebajikan. Penekanan akan diberi ke atas masalah, penilaian dan peningkatan kebajikan haiwan. Pengaruh interaksi manusia-haiwan, domestikasi dan persekitaran fizikal yang disediakan oleh manusia ke atas kelakuan dan kebajikan haiwan dibincangkan. <i>This course covers various aspects of animal behaviour, causes of behaviour and their implications on management, production, health and welfare. Emphasis will be given to the problems, assessment and improvement of animal welfare. The influence of human-animal interaction, domestication and physical environment provided by human beings on behaviour and welfare of animals are discussed.</i>		
SHW3106	Sains Daging / <i>Meat Science</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi prinsip pembentukan dan komposisi daging. Faktor penentuan kualiti daging, kaedah penyimpanan dan pemprosesan daging dan prinsip <i>Hazard Analysis Critical Control Point</i> (HACCP) yang penting untuk kegunaan selepas penyembelihan dan proses hiliran turut dibincangkan. <i>This course incorporates the principles of meat formation and composition. Factors determining meat quality, meat storage and processing methods, and principle of Hazard Analysis Critical Control Point (HACCP) which is important for post-slaughtering and meat processing are also discussed.</i>		

SHW3200	Anatomi dan Fisiologi Haiwan Domestik I / <i>Anatomy and Physiology of Domestic Animals I</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi pengajian tentang bentuk dan fungsi sel, tisu, organ, dan sistem haiwan domestik. Penekanan diberi ke atas perhubungan antara sistem endokrin, reproduksi, laktasi dan saraf. Perbezaan anatomi dan fisiologi pelbagai sistem antara spesies ternakan turut dibincangkan.

This course encompasses the study of form and function of cells, tissues, organs and systems of domestic animals. Emphasis is given to the relationship between the endocrine, reproduction, lactation and nervous systems. The differences in anatomy and physiology of various systems between animal species are also discussed.

SHW3201	Anatomi dan Fisiologi Haiwan Domestik II / <i>Anatomy and Physiology of Domestic Animals II</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi struktur anatomi dan fisiologi haiwan berkenaan sistem pencernaan, rangka otot, kardiovaskular dan urinar. Ia juga meliputi sistem pernafasan haiwan mamalia dan unggas. Termokawalatur dan kesan persekitaran terhadap fungsi fisiologi haiwan turut dibincangkan.

This course encompasses the anatomical structures and physiology in relation to the digestive, musculoskeletal, cardiovascular, and urinary systems. It also covers the respiratory system of the mammals and avian. Thermoregulation and environment's influence on the physiological function of animals are also discussed.

SHW3203	Reproduksi Haiwan / <i>Animal Reproduction</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi proses fisiologi reproduksi haiwan ladang termasuk faktor yang mempengaruhi fertiliti. Pendedahan kepada faktor kesuburan sel pembiakan dan kitaran estrus untuk memanipulasi kaedah pembiakan secara semulajadi dan tiruan. Ia merangkumi penggunaan bioteknologi untuk menambah baik prestasi reproduksi dan meningkatkan produksi ternakan.

This course covers the physiological processes of reproduction in farm animals, including factors influencing fertility. Exposure to factors that influencing the gonad cells and estrus cycle to manipulate the natural and human-assisted reproduction process. It encompasses the use of biotechnology to improve reproductive performance and increase the livestock production.

SHW3303	Genetik Haiwan / <i>Animal Genetics</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini memperkenalkan prinsip asas genetik haiwan. Pendedahan kepada prinsip *Mendel* dan perubahan yang berkaitan kepada hubungan antara alel dan lokus. Penekanan asas pemindahan gen serta ekspresi trait kualitatif. Variasi kromosom, kepelbagaian genetik dan asas molekul serta faktor yang mempengaruhi struktur genetik populasi juga dibincangkan.

This course introduces the basic principles of animal genetics, Mendellian and interaction between alleles and locus. It emphasises on the fundamental of gene transmission and expression of qualitative traits. Chromosomal variation, genetic diversity and basic molecular and factors influencing structure of population genetics are discussed.

SHW3304	Pembiakbakaan Ternakan / <i>Livestock Breeding</i>	3 (2+1)
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Prasyarat : SHW3303

Kursus ini memperkenalkan prinsip asas genetik kuantitatif. Ia menumpukan kepada aplikasi prinsip genetik dalam peningkatan prestasi ternakan. Ia merangkumi kaedah penganggaran dan pemilihan ternakan dalam program pembiakbakaan. Prinsip untuk mengenal pasti ternakan pembiak yang unggul dan meramal respons terhadap pemilihan ditekankan.

This course introduces the basic principles of quantitative genetics. It focuses on the application of genetic principles to improve livestock performance. It includes methods of estimation and selection for animal breeding program. Exposure to Principles to identify superior breeding animals and predict the response to selection are emphasised.

SHW3403	Pemakanan Haiwan Asas / <i>Basic Animal Nutrition</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi klasifikasi, ciri-ciri, fungsi, gejala kekurangan dan metabolisme nutrien. Ini termasuk proses pencernaan, penilaian bahan makanan, piawai pemberian makanan, pengambilan makanan, metabolisme dan keperluan nutrien, pengelasan dan ciri bahan makanan ternakan dan perumusan rangsum haiwan ternakan. <i>This course covers the classification, characteristics, functions, deficiency symptoms and metabolism of nutrients. It includes digestive processes, feed evaluation, feeding standards, feed intake, metabolism and requirement of nutrients, classification and characteristics of feedstuffs and formulation of ration of farm animals.</i>		
SHW3405	Sains Pastura / <i>Pasture Science</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi kepentingan tanaman pastura sebagai sumber makanan haiwan. Ia termasuk pembentukan tanah, sifat kimia, fizik dan biologi tanah, ciri spesies rumput dan kekacang, serta kaedah penubuhan dan pengurusan pastura. Kaedah penggunaan dan penyimpanan pastura turut dibincangkan. <i>This course encompasses the importance of pasture plants as an animal feed source. It includes soil formation, physical, chemical and biological properties of soil, the characteristic of grasses and legumes species, methods of establishment and management of pasture. Methods of utilisation and conservation of pasture are also discussed.</i>		
SHW3407	Biokimia Haiwan / <i>Animal Biochemistry</i>	3 (2+1)
Prasyarat : Tiada Kursus ini memberi pemahaman tentang prinsip biokimia asas dalam organisma hidup khususnya haiwan. Topik pembelajaran merangkumi peranan, fungsi, laluan metabolik, dan kawalan tindak balas biokimia melibatkan biomolekul seperti karbohidrat, asid amino, protein, lipid dan asid nukleik dalam sistem biologi. Integrasi metabolisme di antara tisu haiwan turut dibincangkan. <i>This course provides an understanding on the principle of fundamental biochemistry in living organisms, especially animals. This encompasses the roles, functions, metabolic pathways, and control of the biochemical reactions involving biomolecules such as carbohydrates, amino acids, proteins, lipids and nucleic acids in biological system. Metabolic integrations between animal tissue are also discussed.</i>		
SHW3501	Sains Haiwan Pedaging / <i>Meat Animal Science</i>	3 (2+1)
Prasyarat : SHW3201 Kursus ini meliputi prinsip dan amalan pengurusan pengeluaran haiwan pedaging. Ia merangkumi aplikasi konsep fisiologi, pemakanan, pembiakan, kesihatan gerompok, ekonomi dan pengurusan di bawah sistem penternakan di kawasan tropika yang lembap. <i>This course covers the principles and management practices for meat animal production. It encompasses the application of concepts in physiology, nutrition, breeding, herd health, economics and management for production systems in humid tropics.</i>		
SHW3502	Sains Tenusu / <i>Dairy Science</i>	3 (2+1)
Prasyarat : SHW3201 Kursus ini meliputi pengurusan ternakan tenusu dari lahir hingga dewasa. Penekanan ke atas pengurusan reproduksi, pemakanan dan pemberian makanan, masalah penyakit metabolisme, pengurusan sistem pemerahan susu, pengendalian susu di ladang dan perkembangan industri tenusu. <i>This course covers management of dairy animals from birth until adulthood. Emphasis is given on reproductive management, feed and feeding, problems of metabolic diseases, management of milking systems, handling of milk on the farm and development of the dairy industry.</i>		

SHW3510	Sains Poltri/ <i>Poultry Science</i>	3 (2+1)
Prasyarat : SHW3403 Kursus ini meliputi prinsip produksi daging dan telur poltri komersil. Ia menekankan aspek fisiologi reproduksi dan persekitaran, genetik dan pembiakbakaan, pemakanan, pengurusan am, pemprosesan produk, pengurusan sisa dan kebajikan poltri. Isu di dalam produksi poltri di iklim panas humid tropika turut dibincangkan. <i>This course covers principles of commercial poultry meat and egg production. It emphasizes aspects of reproductive and environmental physiology, genetics and breeding, nutrition, general management, product processing, waste management and welfare of poultry. Issues in poultry production in hot humid tropical climate are also discussed.</i>		
SHW3514	Sains Babi/ <i>Swine Science</i>	3 (2+1)
Prasyarat : SHW3403 Kursus ini meliputi aspek utama pengeluaran babi secara moden. Ianya merangkumi sistem pengeluaran, pengurusan dan amalan umum ladang, pemakanan dan sistem pembiakbakaan. Isu dalam penternakan babi turut dibincangkan. <i>This course covers the main aspects of modern pig production. It encompasses the production systems, general farm management and practices, nutrition, and breeding systems. Issues on pig farming are discussed.</i>		
SHW3602	Statistik Sains Haiwan / <i>Statistics for Animal Science</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep asas dalam statistik, kebarangkalian, frekuensi taburan, statistik lokasi dan serakan. Ia juga meliputi pengumpulan, persembahan dan analisis data, ujian hipotesis dan pengulasan keputusan. Pemilihan kaedah analisis yang bersesuaian untuk data yang berbeza turut dibincangkan. <i>This course encompasses basic concepts in statistics, probability, frequency distribution, statistics of location and dispersion. It also covers data gathering, display and analysis, hypothesis testing and analysis of results. Selection of suitable analysis method for different data is also discussed.</i>		
SHW3700	Pengurusan Sisa Ternakan/ <i>Livestock Waste Management</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi prinsip dan aplikasi pengurusan sisa peternakan dalam sistem intensif. Penekanan diberi kepada sistem dan teknologi di kawasan humid tropika. Kualiti alam sekitar, polisi dan perundangan terhadap pengurusan sisa peternakan dibincangkan. <i>This course encompasses the principles and application of livestock waste management in intensive system. Emphasis is given on systems and technologies in the humid tropics. Environmental quality, policy and legislation on livestock waste management is discussed.</i>		
SHW4100	Hidupan Liar dan Haiwan Rekreasi/ <i>Wildlife and Recreational Animals</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi pelbagai aspek pengurusan hidupan liar dan haiwan rekreasi domestik dalam peliharaan di institusi zoologi, pusat pemulihan dan taman rekreasi. Asas pengurusan spesies haiwan khusus, kaedah pembancian dan konservasi hidupan liar turut dibincangkan. <i>This course encompasses various aspects of managing wild and domestic recreational animals in captivity at zoological institutes, rehabilitation centres and recreational parks. Basic of species-specific animal management, wildlife survey, and conservation are also discussed.</i>		
SHW4106	Mikrobiologi Haiwan / <i>Animal Microbiology</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep asas berkaitan dengan sel-sel prokaryotik, eukaryotik dan virus. Perbincangan berkaitan evolusi dan klasifikasi, ciri-ciri dan perbezaan di antara prokaryotik dan eukaryotik, prinsip am		

pertumbuhan mikrob, ekologi semulajadi mikroorganisma, kebaikan mikroorganisma di dalam produksi haiwan, dan bagaimana mikroorganisma menjadi punca penyakit kepada haiwan ternakan turut diberi penekanan.

This course covers the concepts relating to prokaryotic, eukaryotic cells and viruses. Discussion on evolution and classification, different features between prokaryotic and eukaryotic, the general principles for microbial growth, the natural ecology of microorganisms, the beneficial involvement of microorganisms in animal production, and how microorganisms become the source of disease for livestock are also emphasised.

SHW4201	Fisiologi Persekitaran / <i>Environmental Physiology</i>	3 (3+0)
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Prasyarat : Tiada

Kursus ini meliputi pengajian respons fisiologi oleh haiwan dalam pelbagai iklim persekitaran termasuk cara haiwan menyesuaikan diri terhadap persekitaran mikro dan makro. Penekanan diberi kepada kesan perubahan persekitaran terhadap fungsi fisiologi haiwan seperti dalam sistem saraf, endokrin, pembiakan dan kardiovaskular. Kesan persekitaran terhadap kelakuan dan kesihatan haiwan turut dibincangkan.

This course covers the study of physiological responses by animals in various climatic environments including the ways animals adapt to its micro- and macro environment. Emphasis is given to the effects of the environmental changes on the animal physiological functions such as in nervous, endocrine, reproductive and cardiovascular systems. The environmental effect towards the animal behavior and health are also discussed.

SHW4300	Genetik dan Pembiakbakaan Ternakan Gunaan / <i>Applied Livestock Genetics and Breeding</i>	3 (2+1)
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Prasyarat : SHW3303

Kursus ini meliputi prinsip asas genomik haiwan dan pembiakbakaan ternakan. Kursus ini menekankan aplikasi praktik prinsip-prinsip tersebut dalam analisis genetik populasi dan produksi ternakan. Ianya meliputi kaedah analisis menggunakan teknologi genetik molecular dan teknik pengiraan.

This course covers the basic principles of animal genomics and animal breeding. The course emphasises on the practical application of these principles in population genetic analysis and livestock production. It includes analytical method with the application of molecular genetics data and calculation methods.

SHW4301	Sitogenetik Haiwan / <i>Animal Cytogenetics</i>	3 (2+1)
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Prasyarat : SHW3303

Kursus ini memberi penekanan ke atas morfologi kromosom, proses mitosis dan meiosis, kromosom jantina, penentuan jantina, dan perubahan struktur struktur dan bilangan kromosom. Kepentingan, kegunaan dan implikasi sitogenetik dalam bidang produksi dan pembiakan ternakan dibincangkan.

This course emphasises on chromosome morphology, mitotic and meiotic processes, sex chromosome, sex determination, and structural and numerical chromosomal aberrations. The importance, use and implication of cytogenetics in animal production and breeding are discussed.

SHW4401	Teknologi Pemprosesan Makanan Haiwan / <i>Feed Processing Technology</i>	3 (3+0)
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Prasyarat : SHW3403

Kursus ini meliputi kaedah pemprosesan bahan makanan untuk meningkatkan nilai pemakanan melalui proses fizikal, kimia dan biologi. Teknik pemprosesan dan kesannya terhadap nilai pemakanan dibincangkan. Ianya juga meliputi aspek nutraseutikal.

This course covers methods of feed processing to enhance its nutritive value through physical, chemical, and biological processes. Processing techniques and its effects on the nutritive value are discussed. It also includes the nutraceutical aspect.

SHW4402	Bioteknologi Dalam Makanan Haiwan / <i>Biotechnology in Animal Feed</i>	3 (3+0)
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Prasyarat : SHW3403

Kursus ini meliputi aplikasi bioteknologi dalam makanan dan pemberian makanan haiwan. Penggunaan aditif makanan haiwan, tanaman transgenik, protein dan asid amino, enzim, antibakteria, prebiotik dan probiotik, dan pengawalan pencemaran melalui penggunaan aditif makanan diberi penekanan.

This course covers the application of biotechnology in animal feeds and feeding. The use of feed additives, transgenic plants, proteins and amino acids, enzymes, antibacterials, prebiotics and probiotics, and control of pollution through the use of feed additives are emphasised

SHW4403	Makanan dan Pemberian Makanan / <i>Feeds and Feeding</i>	3 (3+0)
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Prasyarat : SHW3403

Kursus ini merangkumi pengelasan, kualiti, pemprosesan dan penilaian pemakanan bahan makanan, keperluan nutrien untuk saraan dan pengeluaran, dan kaedah perumusan makanan haiwan.

This course incorporates classification, quality, processing and nutritive evaluation of feedstuff, nutrient requirement for maintenance and production, and ration formulation for animals.

SHW4404	Pemakanan Haiwan Gunaan / <i>Applied Animal Nutrition</i>	3 (2+1)
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Prasyarat : SHW3403

Kursus ini meliputi aplikasi lanjutan pemakanan haiwan di dalam produksi ruminan dan bukan ruminan. Penekanan diberi kepada kaedah kajian pemakanan haiwan dan perumusan rangsum yang memenuhi keperluan nutrien haiwan. Konsep pemakanan haiwan persis turut dibincangkan.

This course covers the advanced application of nutrition in ruminant and non-ruminant production. Emphasis is given on the methods of animal nutrition research and ration formulation that meets the nutritional requirement of animals. The concept of precision animal nutrition is discussed.

SHW4500	Rekabentuk dan Infrastruktur Ladang Ternakan / <i>Livestock Farm Design and Infrastructure</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi prinsip pembangunan sebuah ladang ternakan moden dan efisien. Ia meliputi susunatur ladang, infrastruktur asas, perumahan, peralatan dan spesifikasi kejuruteraan dalam rekabentuk dan pembinaan untuk memenuhi keperluan haiwan dan perundangan.

This course encompasses the principles in the development of a modern and efficient livestock farm. It includes farm layout, basic infrastructures, housing, equipment and engineering specifications in the design and construction to meet animals' and regulatory requirements

SHW4501	Produksi Ruminan Kecil / <i>Small Ruminant Production</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi sistem peternakan kambing dan biri-biri termasuk aspek pengurusan, pemakanan, pembiakan, kawalan penyakit dan pemprosesan hasil.

This course encompasses farming systems of goats and sheep which include aspects of management, nutrition, breeding, disease control, and processing of products.

SHW4502	Produksi Tenusu / <i>Dairy Production</i>	3 (2+1)
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Prasyarat : SHW3502

Kursus ini merangkumi pengurusan lanjutan lembu tenusu, teknologi pemerahan susu, penjagaan piawaian kualiti susu. Perbincangan masalah yang sering dihadapi oleh pengusaha turut ditekankan.

This course encompasses advanced dairy cow management, milking technology, maintaining milk quality standard. Discussion on problems commonly faced by entrepreneurs are also emphasised.

SHW4503	Integrasi Ternakan-Tanaman / <i>Livestock-Crop Integration</i>	3(3+0)
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Prasyarat : Tiada

Kursus ini meliputi aspek asas pembangunan dan pelaksanaan sistem penternakan integrasi dengan tanaman perladangan. Penekanan diberi kepada pengeluaran ternakan dalam ladang. Ini termasuk persediaan memenuhi piawai amalan penternakan baik untuk kelestarian sistem. Sumber foraj, pengurusan pemakanan dan produktiviti ternakan turut dibincangkan.

This course covers basic aspects of the development and implementation of livestock integrated farming system with plantation crop. Emphasis is given to livestock production in plantations. This encompasses preparation

to meet standards of good animal husbandry practices for system sustainability. Sources of forages, feeding management, livestock productivity are discussed.

SHW4504	Pengurusan Ragutan / <i>Grazing Management</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini memberi tumpuan kepada jenis dan objektif padang ragut, kesan ragutan ke atas ekosistem, komponen defoliasi, faktor yang mengawal ragutan terpilih, dan prinsip pengurusan kawasan ragutan tradisi dan komersial. Pelajar akan diperkenalkan kepada prinsip asas pengurusan padang ragut semula jadi dan kawasan ragut yang ditambah baik, dan amalan ragut komuniti.

This course focuses on types and objectives of grazing land, effect of grazing on ecosystems, defoliation components, factors governing selective grazing, principles of traditional and commercial grazing management systems. Students will be introduced to basic principles of grazing management in natural and improved grazing lands, and community-based grazing practices.

SHW4505	Logistik Dan Rantaian Bekalan Produk Ternakan / <i>Logistics and Supply Chain of Livestock Products</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini memberi penekanan terhadap prosedur logistik dan rantaian bekalan produk ternakan (daging, susu, telur) dari ladang, semasa pengangkutan dan pengedaran, dan/atau di pasaran bagi tujuan mengekalkan kualiti. Teknik asas penilaian pengurusan yang merangkumi pengendalian pasca tuai, pemprosesan, pembungkusan, penyimpanan, logistik dan pengangkutan akan dibincangkan.

This course focuses on logistics procedures and supply chain of livestock products (meat, milk, eggs) from farm, during transportation and distribution, and/or at the market for quality maintenance. Basic management assessment techniques covering post-harvest handling, processing, packaging, storage, logistics and transportation will be discussed.

SHW4511	Sains Babi Lanjutan / <i>Advanced Swine Science</i>	3 (2+1)
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Prasyarat : SHW3514

Kursus ini meliputi aspek lanjutan dalam produksi babi termasuk reproduksi dan pengurusan dalam penyapihan awal. Kaedah penggunaan dan manipulasi, dan aplikasi bioteknologi ditekankan untuk meningkatkan produktiviti babi turut dibincangkan.

This course covers advanced aspects in pig production including reproduction, management in early weaning. Methods of feed utilization and manipulation, and application of biotechnology to enhance productivity of pigs are also discussed.

SHW4512	Produksi Ayam Penelur / <i>Layer Production</i>	3 (2+1)
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Prasyarat : SHW3510

Kursus ini meliputi pengurusan di ladang ayam penelur. Aspek penting sains poltri dan aplikasi praktikal untuk kejayaan pengurusan operasi diintegrasikan dalam produksi ayam penelur. Penekanan diberi ke atas status industri, baka, susun atur ladang, sistem pemeliharaan, dan kawalan penyakit ayam penelur. Kualiti, kaedah pemprosesan, dan pengurusan sisa telur dibincangkan.

This course covers the management of layer farms. Important aspects of poultry science and their practical application to successfully manage layer operations are integrated into the layer production. This course emphasises the industry status, breed, farm layout, management system, and disease control in layers. Quality, processing methods, and waste management of eggs are discussed.

SHW4513	Produksi Ayam Pedaging / <i>Broiler Production</i>	3 (2+1)
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Prasyarat : SHW3510

Kursus ini meliputi pengurusan dan penternakan ayam pedaging. Aspek penting sains poltri dan aplikasi praktikal untuk kejayaan pengurusan operasi diintegrasikan dalam produksi ayam pedaging. Penekanan diberi ke atas status industri, konsep pengeluaran bersepadu, baka, susun atur ladang, sistem pemeliharaan, kawalan penyakit, dan penuaian ayam pedaging. Pengeluaran, kaedah pemprosesan, dan kualiti daging ayam dibincangkan.

This course covers the management and husbandry of broiler chicken. Important aspects of poultry science and their practical application to successfully manage broiler operations are integrated into the broiler production. This course emphasises the industry status, the concept of integration, breed, farm layout, management system, disease control, and harvesting of broilers. Production, processing methods, and quality of broiler meat are discussed.

SHW4514	Produksi Ayam Pembiak / <i>Poultry Breeder Production</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengurusan dan penternakan ayam pembiak penelur dan pedaging. Aspek penting sains poltri dan aplikasi praktikal untuk kejayaan pengurusan operasi diintegrasikan dalam produksi ayam pembiak. Penekanan diberi ke atas status industri, stok pembiakbakaan, susun atur ladang, sistem pemeliharaan, dan kawalan penyakit ayam pembiak. Teknik pembiakan, penilaian prestasi, dan pengurusan telur bernas ayam pembiak turut dibincangkan.

This course covers the management and husbandry of both layer and broiler breeders. Important aspects of poultry science and their practical application to successfully manage breeder operations are integrated into the poultry breeder production. This course emphasises the industry status, parent stocks, farm layout, management system, and disease control of breeders. Mating techniques performance evaluation and fertile egg management of breeders are discussed.

SHW4515	Pengurusan Burung Walit / <i>Swiftlet Management</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengurusan dan reka bentuk rumah burung walit. Penekanan diberi ke atas status industri, ekologi, spesies, proses pembentukan sarang burung walit, dan amalan semasa burung walit untuk memastikan kejayaan operasi industri burung walit. Penuaian dan penilaian kualiti sarang burung serta cabaran dan isu semasa industri burung walit dibincangkan.

This course covers the management and design of swiftlet houses. This course emphasises the industry status, ecology, species, process of edible bird's nest formation, and current practices in the swiftlet industry to ensure successful operations. Harvesting and evaluation of edible bird's nest quality, as well as the current challenges and issues of the swiftlet industry are discussed.

SHW4540	Pengurusan Ekuin / <i>Equine Management</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi aspek asas dalam penjagaan dan pengurusan ekuin. Ini termasuk makanan dan amalan pemberian pemakanan, penjagaan telapak, kesihatan umum dan kelakuan kuda di stabel.

This course encompasses basic aspects in equine care and management. This includes feeds and feeding practices, hoof care, general health, and behaviour of horses in stable.

SHW4600	Metodologi Penyelidikan / <i>Research Methodology</i>	3 (2+1)
Prasyarat : SHW3602 Kursus ini merangkumi konsep dan teknik asas dalam pengendalian penyelidikan, analisis dan pentafsiran data. Ini termasuk perancangan kajian saintifik, pemilihan kaedah, menganalisis data dan menafsirkan keputusan yang diperolehi serta mempersembahkan hasil kajian. <i>This course covers basic concepts and techniques in conducting research, methodology selection, data analysis and interpretation. It includes planning a scientific research, analysis and interpretation of the results generated and presentation of the findings.</i>		
SHW4703	Teknologi Pemprosesan Daging / <i>Meat Processing Technology</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi aspek teknologi dalam pengendalian dan pemprosesan daging. Ini termasuk penyembelihan, pengendalian, pemprosesan, pengawetan dan pembungkusan. Penilaian dan kawalan kualiti produk daging terproses dibincangkan. <i>This course encompasses technological aspect in meat handling and processing. This includes slaughtering, handling, processing, preservation and packaging. Evaluation and quality control of processed meat products are discussed.</i>		
SHW4801	Perancangan dan Penilaian Sumber / <i>Resource Planning and Evaluation</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi penilaian sumber untuk perancangan dan kemajuan projek haiwan. Kaedah pendekatan sistem yang digunapakai melalui kombinasi komponen dalam sistem pengeluaran haiwan turut dibincangkan. <i>This course encompasses the evaluation of resources for the planning and development of project in animal production. Exposure towards system approach method is utilised through combination of components in animal production systems are discussed.</i>		
SHW4959	Disertasi Bachelo / <i>Final Year Project</i>	6 (0+6)
Prasyarat : Tiada Kursus ini merangkumi pengendalian kajian yang melibatkan perancangan, penggunaan reka bentuk eksperimen dan kaedah yang bersesuaian, pengumpulan dan analisis data, pentafsiran keputusan, pembentangan seminar dan penulisan disertasi. <i>This course involves the conduct of research that includes planning, the use of appropriate experimental design and methods, data collection and analyses, interpretation of results, seminar presentation and dissertation writing.</i>		
SHW4901	Latihan Industri / <i>Industrial Training</i>	12 (0+12)
Prasyarat : Tiada Kursus ini melibatkan penempatan pelajar selama 24 minggu di sektor industri ruminan dan bukan ruminan. Pelajar memperolehi pengalaman mengurus serta berupaya melapor keberkesanan amalan kerja. <i>This course involves student's attachment for 24 weeks at ruminant and non-ruminant industrial sectors. Students can obtain management experience and to report the effectiveness of work practices.</i>		

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Sains Tanaman Department of Crop Science

AGR3507	Tanaman Makanan Utama/ <i>Main Food Crops</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pelbagai sistem dan teknik penanaman dan penyelenggaraan tanaman makanan utama termasuk bijirin, tanaman umbisi dan sayuran. Pengurusan lepas tuai dan amalan lestari dibincangkan.

This course covers various planting systems and techniques and maintenance of main food crops including cereals, legumes, and vegetables. Post-harvest management and sustainable practices are also discussed.

AGR3508	Tanaman Perladangan Utama/ <i>Main Plantation Crops</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi prinsip pengeluaran, pengurusan dan ekonomi komoditi perladangan utama negara. Sistem pengurusan tanaman perladangan menggunakan teknologi terkini dan secara lestari turut dibincangkan.

This course encompasses principles of production, management and economics of nation's main plantation commodities. Plantation crop management systems using the latest and sustainable technologies are also discussed

AGR3001	Amalan Ladang: Tanaman Makanan/ <i>Field Practice Food Crops</i>	1 (0+1)
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Prasyarat : Tiada

Kursus ini memberi penekanan kepada amalan pertanian baik dalam pengeluaran dan pengendalian lepas tuai hasil tanaman makanan seperti padi, jagung, buah-buahan, sayur-sayuran dan lain-lain.

This course emphasizes on good agricultural practice in production and postharvest handling of food crops such as paddy, maize, crops, vegetables and others.

AGR3101	Botani Pertanian/ <i>Agriculture Botany</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi taksonomi, morfologi dan anatomi tanaman pertanian. Asas reproduksi tumbuhan, ekologi tanaman serta prinsip pemeliharaan dan pemuliharaan tumbuhan turut dibincangkan.

This course covers taxonomy, morphology and anatomy of agricultural crops. The basics of plant reproduction, crop ecology as well as principles of plant conservation and preservation are also discussed.

AGR3102	Prinsip Sains Rumpai/ <i>Principles Of Weed Science</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi prinsip, konsep serta ekologi rumpai yang merangkumi klasifikasi, pertapakan dan kesesuaian habitat serta strategi mengawal dan mengurus rumpai di kawasan pertanian dan bukan pertanian.

This course covers principles, concepts and ecology of weed which includes identification, classification, establishment and habitat requirement, and control and management strategies of weeds in agricultural and non-agricultural areas.

AGR3204 Pembiakbakaan Tumbuhan / <i>Plant Breeding</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi sistem pembiakan tumbuhan, pewarisan sifat kuantitatif dan kualitatif, heritabiliti, kaedah pemilihan tanaman terdebunga sendiri dan kacuk, kaedah pembiakbakaan melalui kacukan, mutasi, kacukan antara individu berpertalian genetik jauh, dan kaedah pembaikbakaan yang lain.

The course covers plant reproduction system, inheritance of qualitative and quantitative characters, heritability, selection methods for self- and cross-pollinated crops, breeding through hybridization, mutation, wide hybridization, and other breeding methods.

AGR3301 Fisiologi Tanaman / <i>Crop Physiology</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini merangkumi proses dan fungsi tanaman pada peringkat sel, sub-sel dan keseluruhan tumbuhan sebagai tindak balas kepada pembolehubah dan pertumbuhan persekitaran. Pemahaman tentang fotosintesis, respirasi, transpirasi, translokasi, pengambilan nutrien dan air, pengawal selia pertumbuhan tumbuhan dan proses lain yang mempunyai kesan mendalam terhadap hasil tanaman dibincangkan.

This course covers the processes and functions of the crops at cellular, sub-cellular and whole plant levels in response to environmental variables and growth. Understanding of photosynthesis, respiration, transpiration, translocation, nutrient and water uptake, plant growth regulator and other processes which have profound impact on crop yield are discussed.

AGR3504 Tanaman Industri Minor / <i>Minor Industrial Crops</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi trend dan perkembangan tanaman industri minor. Prinsip pengeluaran, pengurusan dan pemprosesan tanaman turut dibincangkan.

This course covers the trends and development of minor industrial crops. The principles of production, management and processing of those crops are also discussed.

AGR3701 Eksperimentasi dan Biometri Pertanian / <i>Eksperimentation and Agriculture Biometry</i>	4(3+1)
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Prasyarat : Tiada

Kursus ini memberi kefahaman dan keupayaan kepada pelajar untuk menggunakan kaedah statistik dalam penyelidikan. Pemilihan rekabentuk eksperimen, analisis data dan membuat rumusan daripada keputusan eksperimen dalam bidang pertanian dibincangkan.

This course provides students with an understanding and ability to use statistical methods in research. Selection of experimental designs, data analysis and interpretation of results in agricultural experiment are discussed.

AGR4100	Agrobiodiversiti/ <i>Agrobiodiversity</i>	3(3+0)
Prasyarat : Tiada Kursus ini merangkumi aspek biodiversiti, pengurusan dan pengeluaran tanaman baharu dan berpotensi untuk aktiviti pertanian. Kaedah manipulasi persekitaran untuk pengeluaran dan potensi komersil tanaman baharu dibincangkan. <i>This course covers the aspects of biodiversity, management and production of emerging and potential plants for agriculture activities. Methods of environmental manipulation for production and the commercial potential of new crops are discussed.</i>		
AGR4101	Botani Pertanian Lanjutan/ <i>Advanced Agricultural Botany</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi morfologi dan anatomi struktur tumbuhan berdasarkan peringkat pembentukan (morfogenesis) menggunakan kaedah yang bersesuaian. Taksonomi dan sistematik tumbuhan, biologi pendebungaan, ekologi tanaman dan kepelbagaian tumbuhan turut dibincangkan. <i>This course covers the morphology and anatomy of plant structures according to developmental stages (plant morphogenesis) using suitable methods. Plant taxonomy and systematic, pollination biology, crop ecology and plant diversity are discussed.</i>		
AGR4103	Pengurusan Rumpai Perladangan/ <i>Plantation Weed Management</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi klasifikasi, taburan dan kaedah pengurusan rumpai di kawasan perladangan. Penilaian keberkesanan setiap kaedah pengurusan dan impaknya ke atas agro-ekosistem turut dibincangkan. <i>This course covers classification, distribution and management of weeds in plantations. Evaluation on the effectiveness of the management techniques and their impact on the agro-ecosystem is also discussed.</i>		
AGR4203	Genetik Kuantitatif/ <i>Quantitative Genetics</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi genetik populasi dan kuantitatif tumbuhan yang melibatkan perubahan frekuensi gen dan genotip dalam populasi besar dan kecil. Konsep heritabiliti, pemilihan dan kemajuan genetik, analisis keupayaan bergabung, rekabentuk pengawanan dan pemilihan berulang serta analisis lokus sifat kuantitatif (QTL) dan pemetaannya dibincangkan. <i>This course covers population and quantitative genetic involving changes in gene and genotype frequencies in large and small populations. Concept of heritability, selection and genetic advance, analysis of combining ability, mating designs, recurrent selection, concept, and mapping of quantitative trait loci (QTL) will be discussed.</i>		
AGR4301	Fisiologi Tanaman Lanjutan/ <i>Advanced Crop Physiology</i>	3(2+1)
Prasyarat : Tiada Kursus ini meliputi kesan tegasan persekitaran terhadap proses fisiologi, respirasi dan translokasi. Metabolisma pemakanan tanaman meliputi penyerapan dan pengangkutan serta fisiologi tanaman komoditi terpilih dibincangkan.		

This course covers effects of environmental stresses on photosynthesis respiration and translocation. Mineral nutrition metabolism including absorption and translocation and the physiology of selected commodity are discussed.

AGR4302 Sains Racun Herba / <i>Herbicide Science</i>	3(2+1)
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Prasyarat : AGR3102

Kursus ini merangkumi ciri fiziko-kimia, degradasi dan kesan sisa racun herba dan tindak balas tumbuhan terhadap racun herba. Faktor yang mempengaruhi keberkesanan racun herba juga dibincangkan.

This course covers physico-chemical characteristics, degradation and residual effects of herbicide and plant response to herbicide. Factors affecting efficacy of herbicide are discussed.

AGR4303 Pemakanan Tanaman / <i>Crop Nutrition</i>	3(2+1)
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Prasyarat : AGR3102

Kursus ini merangkumi peranan dan fungsi elemen nutrien utama, penyerapan, metabolisme, interaksi antara nutrien dan keupayaan media untuk membekal nutrien. Kesan kekurangan dan keterlembhan nutrient kepada tumbesaran, penghasilan tanaman dan alam sekitar dibincangkan.

This course encompasses the role and function of major nutrient elements, absorption, metabolism, interactions amongst nutrients and the capacity of media to supply nutrients. Effects of nutrient deficiency and excess on growth, yield and the environment are discussed.

AGR4501 Pengurusan Pastura / <i>Pasture Management</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penilaian spesies rumput dan kekacang pastura tropika. Penanaman dan pengurusan pastura untuk membekalkan makanan dalam pelbagai sistem pengeluaran ternakan ruminan turut dibincangkan.

This course encompasses an evaluation of tropical pasture grass and legume species. Planting and pasture management to provide feed in various ruminant production systems is also discussed.

AGR4503 Pengeluaran Tanaman Padi / <i>Rice Crop Production</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penilaian spesies rumput dan kekacang pastura tropika. Penanaman dan pengurusan pastura untuk membekalkan makanan dalam pelbagai sistem pengeluaran ternakan ruminan turut dibincangkan.

This course encompasses an evaluation of tropical pasture grass and legume species. Planting and pasture management to provide feed in various ruminant production systems is also discussed.

AGR4507 Pengeluaran Tanaman Kelapa Sawit / <i>Oil Palm Crop Production</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengeluaran, pengurusan dan pemprosesan hasil industri tanaman kelapa sawit. Faktor yang mempengaruhi keberkesanan pengeluaran perladangan kelapa sawit dibincangkan.

This course covers the production, management and processing of oil palm. Factors affecting the efficiency of production and management of oil palm plantation are discussed.

AGR4507	Pengeluaran Tanaman Kelapa Sawit / <i>Oil Palm Crop Production</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengeluaran, pengurusan dan pemprosesan hasil industri tanaman kelapa sawit. Faktor yang mempengaruhi keberkesanan pengeluaran perladangan kelapa sawit dibincangkan.

This course covers the production, management and processing of oil palm. Factors affecting the efficiency of production and management of oil palm plantation are discussed.

AGR4507	Pengeluaran Tanaman Kelapa Sawit / <i>Oil Palm Crop Production</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengeluaran, pengurusan dan pemprosesan hasil industri tanaman kelapa sawit. Faktor yang mempengaruhi keberkesanan pengeluaran perladangan kelapa sawit dibincangkan.

This course covers the production, management and processing of oil palm. Factors affecting the efficiency of production and management of oil palm plantation are discussed.

AGR4508	Pengeluaran Tanaman Getah / <i>Rubber Crop Production</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengeluaran, pengurusan dan pemprosesan getah. Faktor yang mempengaruhi keberkesanan pengeluaran dan pengurusan perladangan getah dibincangkan.

This course covers production, management and processing in rubber industry. Factors affecting the efficiency of production and management of estates are discussed.

AGR4511	Pengurusan Perladangan Getah / <i>Rubber Plantation Management</i>	3(1+2)
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Prasyarat : Tiada

Kursus ini meliputi industri getah termasuk pengeluaran, pengurusan dan pemprosesan hasil. Faktor yang mempengaruhi keberkesanan pengeluaran perladangan getah juga dibincangkan.

This course covers the rubber industry including production, management and processing. Factors affecting the efficiency of rubber plantation production are also discussed.

AGR4801	Pertanian Lestari / <i>Sustainable Agriculture</i>	2(2+0)
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Prasyarat : Tiada

Kursus ini merangkumi pendekatan secara holistik untuk membangunkan projek pertanian secara lestari berdasarkan faktor fungsian agro-ekosistem. Kaedah penilaian kelestarian sesuatu sistem pembangunan pertanian dibincangkan.

This course covers holistic approaches to develop a sustainable agricultural project based on the functional factors of agro-ecosystem. Methods to evaluate the sustainability of agricultural systems are discussed.

AGR4802	Teknologi Biji Benih / <i>Seed Technology</i>	3(2+1)
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Prasyarat : AGR3301 atau HRT3002

Kursus ini merangkumi aspek pembentukan, pemprosesan dan kualiti biji benih. Faktor yang mempengaruhi kualiti biji benih dan persekitaran penyimpanan bagi pengeluaran biji benih komersil dibincangkan.

This course encompasses the aspect of seed formation, processing and quality of seeds. Factors influencing seed quality and storage environment for commercial seed production are discussed.

HRT3403	Pengeluaran Tanaman Hortikultur / <i>Horticulture Crop Production</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengelasan, pembiakan, pengeluaran dan penyelenggaraan tanaman hortikultur. Pengurusan dan pengendalian lepas tuai tanaman turut ditekankan.

This course covers classification, propagation, production and maintenance of horticultural crops and postharvest management and handling are also emphasized.

HRT3301	Hortikultur Hiasan / <i>Ornamental Horticulture</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi klasifikasi, prinsip dan teknik dalam pengeluaran tanaman hiasan yang merangkumi pengecaman, penanaman, penyelenggaraan dan pengurusan.

This course covers classifications, principles and techniques of ornamental plant production, which include identification, planting, maintenance and management.

HRT3442	Sains Turf / <i>Science Turf</i>	2(1+1)
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Prasyarat : Tiada

Kursus ini meliputi pengelasan dan pengecaman spesies rumput turf serta kegunaannya. Keperluan pertumbuhan, pembiakan, penanaman dan pengurusan rumput turf dibincangkan.

This course covers classification and identification of turfgrass species and their uses. Growth requirements, propagation, establishment, and management of turfgrass are discussed.

HRT3412	Olerikultur / <i>Olericulture</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengelasan, prinsip dan teknik pengeluaran di lapangan dan persekitaran terkawal. Faktor pengeluaran, pembiakan, penanaman, pengurusan dan pengendalian lepas tuai tanaman sayur-sayuran juga diberi penekanan.

This course covers classification, principles and production techniques in the field and controlled environment. Factors of production, breeding, planting, management and handling of vegetable crops are also emphasized.

HRT3422	Pomologi / <i>Pomology</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengelasan dan teknik pengeluaran buahbuahan di lapangan. Pembiakan, amalan kultura dan pengendalian lepas tuai tanaman buah-buahan dibincangkan.

This course covers the classification and techniques of fruit production in the field. Breeding, cultural practices and postharvest handling of fruit crops are discussed.

HRT3432	Hortikultur Hiasan / <i>Ornamental Horticulture</i>	2(1+1)
Prasyarat : Tiada Kursus ini meliputi pelbagai jenis tanaman hiasan yang diaplikasi ke ruang landskap. Keperluan tanaman dan penyelenggaraan yang meliputi pengairan, pembajaan, pemangkasan dan pengendalian lepas tuaidibincangkan. <i>This course covers various types of ornamental plants that are applied to landscape spaces. Crop requirements and maintenance which includes irrigation, fertilization, pruning handling are discussed.</i>		
HRT3600	Fisiologi Lepas Tuai / <i>Postharvest Physiology</i>	3(2+1)
Prasyarat : Tiada Kursus ini meliputi prinsip dan teknik pengendalian, penyimpanan, penilaian kualiti dan keselamatan komoditi hortikultur. Mekanisme fisiologi kematangan, kemasakan dan senesens dibincangkan. <i>This course covers the principles and techniques of handling, storage, quality evaluation and safety of horticultural commodities. Physiological mechanisms related to maturation, ripening and senescence are discussed.</i>		
HRT3800	Pengenalan Rekabentuk Landskap / <i>Introduction to Landscape Design</i>	3(2+1)
Prasyarat : HRT3430 dan HRT3431, atau HRT3432 Kursus ini meliputi pelbagai elemen rekabentuk, prinsip dan proses merekabentuk pelan landskap dengan menggunakan bahan lembut dan kejur. Isu semasa berkaitan pengurusan dan penyelenggaraan dibincangkan. <i>This course covers various aspects of design elements, principles and landscape designing processes using soft and hardscape materials. Current issues in landscape management and maintenance are discussed.</i>		
HRT4422	Pengeluaran Komersial Buah-buahan / <i>Commercial Fruit Production</i>	3(2+1)
Prasyarat : Tiada Kursus in merangkumi kriteria, isu dan aspek teknologi pengeluaran buah-buahan bernilai tinggi yang mempunyai impak ekonomi. Prosedur operasi piawaian dan aspek kewangan dalam pengeluaran dan pengurusan dibincangkan. <i>This course covers criterias, issues and aspects of production technology for high economic value fruits. Standard operation procedure and financial aspect in production and management are discussed</i>		
HRT4432	Kultur Orkid / <i>Orchid Culture</i>	3(2+1)
Prasyarat : Tiada Kursus ini meliputi klasifikasi dan pencaman genera, penghibridan dan pewarisan, sistem pembiakan, faktor mempengaruhi tumbesaran dan perkembangan, sistem penanaman, pengurusan dan pengendalian pasca tuai dan pemasaran orkid. <i>This course covers classification and identification of genera, hybridization and inheritance, propagation systems, factors affecting growth and development, planting systems, management and postharvest handling and marketing of orchids.</i>		

HRT4300	Pengawalatur Tumbesaran Tumbuhan / <i>Plant Growth Regulators</i>	3(2+1)
Prasyarat : AGR3301 atau HRT3002 Kursus ini meliputi sintesis, mekanisme dan peranan pengawalatur dalam tumbesaran tanaman. Manipulasi dan kesan pengawalatur tumbesaran terhadap inisiasi bunga, kejadian dan keguguran buah, pasca tuai dan alam sekitar dibincangkan. <i>This course covers classification, synthesis, mechanism and roles of growth regulators in plant growth. Manipulation and effects of growth regulator on flower initiation, fruit set and abscission, postharvest and environment are discussed.</i>		
HRT4400	Tanaman Eksotika / <i>Exotic Crop</i>	2(2+0)
Prasyarat : Tiada Kursus ini merangkumi aspek biodiversiti, pengurusan dan pengeluaran tanaman eksotik. Pemeliharaan, pemuliharaan dan potensi industri tanaman eksotik dibincangkan. <i>This course covers aspects of biodiversity, management and production of exotic plants. Preservation, conservation and industrial potential of exotic plants are discussed.</i>		
HRT4402	Pertanian Bandar / <i>Urban Agriculture</i>	2(2+0)
Prasyarat : Tiada Kursus ini merangkumi aspek perkembangan pertanian bandar serta sumbangannya terhadap sosio-ekonomi dan persekitaran bandar. Pelbagai sistem pengeluaran, pengurusan dan teknologi yang boleh menyumbang kepada amalan pertanian bandar lestari dibincangkan. <i>This course covers aspects of urban agriculture and its contribution to socio-economic and urban environment. Various production systems, management and technology that will contribute to a sustainable production practices are discussed.</i>		
HRT4441	Pengurusan Rumput Turf / <i>Turfgrass Management</i>	3(2+1)
Prasyarat : Tiada Kursus ini merangkumi isu semasa industri rumput turf, pengaruh alam sekitar ke atas tumbesaran, pembiakan, perancangan, penubuhan dan pengurusan rumput turf untuk pelbagai kegunaan. <i>This course encompasses current issues of turfgrass industry, influence of environment on growth, propagation, planning, establishment and management of turfgrass for various uses.</i>		
HRT4800	Kultur Tanpa Tanah / <i>Soilless Culture</i>	3(2+1)
Prasyarat : Tiada Kursus ini meliputi prinsip, teknik dan pengurusan sistem kultur tanpa tanah. Formulasi nutrien, keperluan fizikal dan kultura penanaman pelbagai sistem pengeluaran komersial bagi tanaman hortikultur dibincangkan. <i>This course covers principles, techniques and management of soilless culture system. Nutrient formulation, physical and cultural requirements of various commercial production systems for horticultural crops are discussed.</i>		

HRT4802	Hortikultur Persekitaran Terkawal/ <i>Controlled Environment Horticulture</i>	2(2+0)
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Prasyarat : HRT3000 atau HRT3400 atau AGR3301

Kursus ini merangkumi konsep pengawalan dan manipulasi faktor persekitaran, peralatan, keperluan dan pengurusan tanaman untuk pelbagai sistem tanaman persekitaran terkawal. Potensi pengkomersilan pengeluaran tanaman persekitaran terkawal dibincangkan.

This course covers concept of control and manipulation of environmental factors, instrumentation, crop requirements and management of various cropping systems under controlled environment. Commercialization of potential crop production under controlled environment is discussed.

HRT4803	Pembiakan Mikro Tumbuhan/ <i>Plant Micropropagation</i>	3(2+1)
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Prasyarat : Tiada

Kursus ini merangkumi prinsip, pemilihan eksplan, proses teknik pembiakan, pengurusan dan aplikasi tanaman secara in vitro. Aspek fisiologi pertumbuhan dan ekonomi pengeluaran anak pokok secara komersial dibincangkan.

This course encompasses systems, techniques, management and application of in vitro propagation of crop. Growth physiology and economic aspects in commercial production of seedlings are discussed.

HRT4804	Tumbuhan Ubatan Dan Herba/ <i>Medicinal Plants and Herbs</i>	2(2+0)
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Prasyarat : Tiada

Kursus ini meliputi klasifikasi, prinsip, teknik pengeluaran lapangan dan persekitaran terkawal dengan tumpuan kepada faktor pengeluaran, pembiakan, penanaman, pengurusan dan pengendalian, pasca tuai tanaman ubatan dan herba.

This course covers classifications, principles, field and protected environmental production techniques with emphasis on production factors, propagation, planting, management and postharvest handling of medicinal plants and herbs

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Perlindungan Tumbuhan Department of Plant Protection

PLP3002	Pengurusan Perosak Tanaman Hortikultur / <i>Pest Management of Horticultural Crops</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pengesanan, pengenalpastian dan pemantauan perosak tanaman hortikultur dan turf serta strategi pengurusanannya.

This course covers detection, identification and monitoring pests of horticultural crops and turf as well as strategies for their management.

PLP3003	Perlindungan Tumbuhan Asas / <i>Fundamentals of Plant Protection</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi teknik penyediaan specimen, pengkelasan dan pengecaman perosak (serangga, patogen dan rumpai) serta amalan kawalan dan pengurusan perosak.

This course covers detection, identification and monitoring pests of horticultural crops and turf as well as strategies for their management

PLP3104	Entomologi Asas / <i>Fundamentals Of Entomology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi taksonomi, biologi, kaedah kawalan, pengenalpastian perosak utama tanaman.

Pelaksanaan kawalan serangga dalam konteks pengurusan tanaman bersepadu turut ditekankan.

This course covers taxonomy, biology, methods of control, identification of important crop pests.

Implementation of insect control in the context of integrated crop management is also emphasized.

PLP3105	Pengurusan Perosak Bandaran / <i>Urban Pest Management</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi pengenalan terhadap perosak di kawasan perbandaran meliputi perosak vertebrat dan invertebrat. Kaedah kawalan bermula daripada pengesanan dan pengenalpastian perosak, pengurusan termasuk kawalan biorasional dan keberkesanannya turut ditekankan.

This course covers the introduction of urban pests which include both vertebrate and invertebrate pests. Control methods starting from detection and identification of the pests, management including biorational control and its effectiveness will be emphasized.

PLP3106	Entomologi Tanaman Hortikultur / <i>Horticultural Plant Entomology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi pembezaan simptom kerosakan oleh serangga perosak hortikultur dan hubungkait maklumat kerosakan dengan spesies serangga perosak. Pengurusan serangga perosak secara mesra alam dibincangkan

This course covers differentiation of damage symptoms by various horticultural insect pests. Environmental friendly insect pest management is discussed

PLP3204	Patologi Tumbuhan Asas/ <i>Fundamentals Of Plant Pathology</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi jenis patogen tumbuhan dan penyakit yang disebabkan. Diagnosis dan epidemiologi penyakit serta kaedah kawalan yang lestari dibincangkan. <i>This course encompasses types of plant pathogens and the diseases caused. Disease diagnosis and epidemiology as well as sustainable control techniques are discussed.</i>		
PLP3205	Patologi Tanaman Hortikultur/ <i>Horticultural Plant Pathology</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi prinsip asas patologi tumbuhan dan pengurusan penyakit tanaman hortikultur. Pencaman penyakit utama tanaman hortikultur serta kaedah amalan kawalan penyakit dibincangkan. <i>This course covers the basic principles of plant pathology and disease management of horticultural crops. Identification of important diseases and their control practices are discussed</i>		
PLP4001	Pengawalan Perosak/ <i>Pest Control</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi penilaian dan pengukuran kerosakan dan kerugian infestasi perosak. Pengurusan kawalan yang mesra alam dibincangkan. <u><i>This course covers the evaluation and measurement of damages and losses from pest and disease infestation. Environmentally friendly control management is discussed.</i></u>		
PLP4002	Teknik Penggunaan Racun Perosak/ <i>Pesticide Application Technique</i>	3 (2+1)
Prasyarat : Tiada		
<u><i>Kursus ini merangkumi prinsip utama aplikasi racun perosak, rekabentuk, pemilihan dan penyelenggaraan alatan semburan. Kaedah penggunaan racun perosak yang selamat turut dibincangkan.</i></u> <u><i>This course covers main principle of pesticide application, design, selection and maintenance of sprayer equipments. Safe methods of pesticide application are discussed.</i></u>		
PLP4003	Kawalan Biologi Perosak Tanaman/ <i>Biological Control Of Crop Pests</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi teknik pengawalan perosak tanaman menggunakan musuh semulajadi termasuk pembentukan dan penyampaian produk dan agen kawalan biologi. <i>This course covers techniques of controlling crop pests utilizing natural enemies including the development and delivery of products and bio-control agents.</i>		
PLP4004	Pengurusan Perosak Tanaman Perladangan/ <i>Pest Management of Plantation Crops</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi aspek biologi, ekologi dan epidemiologi perosak dan penyakit utama tanaman perladangan penting di Malaysia dan negara Asia Tenggara yang lain. Penilaian kritis tentang pengurusan penyakit dan perosak berkenaan menggunakan kaedah perundangan, kultur, koresistensi perumah, biologi dan kimia adalah ditekankan. <i>The course covers aspects of biology, ecology and epidemiology of the major pests and diseases of important plantation crops in Malaysia and other Southeast Asian countries. Critical evaluation of the management of these pests and diseases using regulatory, cultural, host resistance, biological and chemical methods is emphasized.</i>		

PLP4005	Pengurusan Perosak Rumput Turf Dan Tanaman Hortikultur/ Pest <i>Management of Turfgrass and Horticultural Crops</i>	3 (2+1)
Prasyarat : PLP3204 Kursus ini meliputi pengesanan, pengenalpastian dan pemantauan perosak tanaman hortikultur dan turf serta strategi pengurusannya <i>This course covers detection, identification and monitoring pests of horticultural crops and turf as well as strategies for their management</i>		
PLP4101	Entomologi/ <i>Entomology</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi peranan serangga sebagai perosak dan agen kawalan biologi. Strategi bersistematik dalam implementasi pengurusan perosak bersepadu dan pengawalaturan populasi serangga dibincangkan. <i>This course covers the role of insects as pests and biological control agents. Systematic strategies in integrated pest management implementation and insect population regulation are discussed.</i>		
PLP4102	Apikultur/ <i>Apiculture</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi arah-aliran evolusi lebah, biologi, ekologi, prinsip dan teknik pengurusan perlembahan di tropika, hasil sampingan perlembahan dan kegunaannya dalam perkembangan perlembahan di rantau ini. <i>This course encompasses the evolutionary trends of honeybees, biology, ecology, principles and techniques in beekeeping management in the tropics, by-products of beekeeping and their uses on the development of beekeeping in the region.</i>		
PLP4201	Patologi Tumbuhan/ <i>Plant Pathology</i>	3 (2+1)
Prasyarat : Tiada <u>Kursus ini merangkumi proses jangkitan, interaksi perumah- patogen epidemiologi penyakit dan mekanisme kerintangan. Pengenalpastian patogen penyakit tanaman dan kaedah kawalan penyakit secara lestari dibincangkan.</u> <u><i>This course encompasses infection processes, plant- pathogen interactions, plant disease epidemiology and mechanisms of resistance. Identification of pathogens and sustainable control methods are discussed.</i></u>		
PLP4601	Sains Racun Perosak/ <i>Pesticide Science</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi ciri utama racun perosak, mod tindakan, keresistanan, penggunaan agen biologi, formulasi dan nasib racun dalam alam sekitar. Teknologi aplikasi racun perosak, kaedah mengkaji dan mengukur tindak balas perosak terhadap racun perosak dibincangkan. <i>This course covers the main characteristics of pesticide mode of action, resistance, use of biological agents, formulations and fate of poison in the environment. Pesticide application technology, methods of studying and measuring pest response to pesticide are discussed</i>		

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Perniagaantani dan Ekonomi Biosumber Department of Agribusiness and Bioresource Economics

PPT3001	Pengenalan kepada Sistem Perniagaantani Malaysia/ <i>Introduction To Malaysia Agribusiness System</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi pemahaman komprehensif mengenai perniagaantani, kepentingan dan hubungannya dengan sektor ekonomi dan peluang pekerjaan di dalam bidang perniagaantani di Malaysia. Hubung kait antara rantai bekalan perniagaantani dengan peranan pengguna, kerajaan, dan institusi kewangan turut dibincangkan. <i>This course encompasses a comprehensive understanding of agribusiness, its importance and relationship to the economy and employment opportunities in the field of agribusiness in Malaysia. The relationship between agribusiness supply chain with the consumer, government, financial institutions roles are also discussed.</i>		
PPT3201	Prinsip Pengurusan Perniagaantani/ <i>Principles of Agribusiness Management</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi prinsip dan fungsi pengurusan firma perniagaantani. Pelbagai kaedah analisis teknikal dan strategi pengurusan perniagaantani yang bersesuaian digunakan dalam membuat keputusan untuk mengatasi masalah pengurusan perniagaantani turut dibincangkan. <i>This course encompasses the principles and management functions of agribusiness firms. Various methods of technical analysis and agribusiness management strategies that are suitable for use in decision making to overcome agribusiness management problems are also discussed.</i>		
PPT3301	Prinsip Pengurusan dan Pemasaran Pertanian/ <i>Principles of Agricultural Management and Marketing</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi konsep asas pengurusan dan pemasaran pertanian, persekitaran pasaran serta kesan dasar kerajaan terhadap perniagaantani. <i>This course encompasses basic concepts of agricultural marketing and management, marketing environment and influence of government policies on agribusiness.</i>		
PPT3401	Perakaunan Perniagaantani/ <i>Agribusiness Accounting</i>	4 (3+1)
Prasyarat : Tiada Kursus ini merangkumi proses mengenal pasti, merekod dan meringkaskan data dan maklumat untuk menyediakan penyata kewangan seperti kunci kira-kira dan penyata pendapatan. Penggunaan maklumat kewangan dan perakaunan di dalam proses membuat keputusan perniagaantani juga dibincangkan. <i>This course covers the process of identifying, recording and summarizing data and information for the preparation of financial statements such as balance sheet and income statement. The use of financial and accounting information in agribusiness decision-making are also discussed.</i>		

PPT3403	Perakaunan Kos Dan Pengurusan Perniagaantani / <i>Agribusiness Cost and Management Accounting</i>	3 (3+0)
Prasyarat : PPT3401 Kursus ini memberi penekanan terhadap aspek perancangan, kawalan, penilaian prestasi dan pembuatan keputusan di dalam sesebuah syarikat perniagaantani dengan menggunakan maklumat perakaunan. Aspek teori dan praktikal dalam pengurusan kos perakaunan bagi memantau prestasi perniagaantani juga diberi penekanan. <i>This course emphasizes on planning, controlling, performance appraisal and decision making in an agribusiness firm by using accounting information. Theoretical and practical aspects in accounting cost management to monitor the performance of agribusiness are also emphasized.</i>		
PPT3702	Keusahawanan Pertanian / <i>Agricultural Entrepreneurship</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep keusahawanan, ciri usahawan, penyediaan pelan perniagaan, pelaksanaan projek, pemasaran, kredit, strategi dan pengurusan sumber serta isu semasa mengenai keusahawanan dalam pertanian. Peranan kerajaan dalam pembangunan keusahawanan pertanian turut dibincangkan. <i>This course covers the concepts of entrepreneurship, the characteristics of an entrepreneur, preparation of a business plan, implementation of project marketing, credit, business strategies, management of resources and current issues of entrepreneurship in agriculture. The role of government in the development of agricultural entrepreneurship are also emphasised.</i>		
PPT3703	Keusahawanan Pertanian / <i>Agricultural Entrepreneurship</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep keusahawanan, ciri usahawan, penyediaan pelan perniagaan, pelaksanaan projek, pemasaran, kredit, strategi dan pengurusan sumber serta isu semasa mengenai keusahawanan dalam pertanian. Peranan kerajaan dalam pembangunan keusahawanan pertanian turut dibincangkan. <i>This course covers the concepts of entrepreneurship, the characteristics of an entrepreneur, preparation of a business plan, implementation of project marketing, credit, business strategies, management of resources and current issues of entrepreneurship in agriculture. The role of government in the development of agricultural entrepreneurship are also emphasized.</i>		
PPT3801	Statistik Perniagaantani / <i>Agribusiness Statistics</i>	3 (2+1)
Prasyarat : Tiada Kursus ini berkaitan penggunaan kaedah statistik di dalam bidang perniagaantani. Ia meliputi konsep dan prinsip analisis statistik yang asas untuk menguji hipotesis dalam kaedah berparameter dan tidak berparameter bagi membuat keputusan dan mencadangkan penyelesaian masalah perniagaantani. <i>This course is related to the usage of statistics method in agribusiness. It covers the basic concepts and principles of statistical analysis for testing hypotheses in parametric and non-parametric methods for decision making and proposing solutions to agribusiness problems.</i>		
PPT4204	Ekonomi Perniagaantani / <i>Agribusiness Economics</i>	3 (3+0)
Prasyarat : EPT3104 Kursus ini meliputi aplikasi teori dan analisis permintaan dan penawaran, keanjalan, kegagalan pasaran, struktur pasaran, pasaran faktor dan persekitaran ekonomi makro dalam membuat keputusan perniagaantani. Dasar kerajaan dan isu ekonomi global yang mempengaruhi prestasi perniagaantani juga diberi penekanan. <i>This course covers the applications of demand and supply theories and analyses, elasticity, market failure, market structure, factor markets and macroeconomic environment in agribusiness decision making. Government policies and global economic issues that affect the performance of agribusiness also emphasized.</i>		

PPT4206	Pengurusan Koperasi Pertanian / <i>Management of Agricultural Cooperative</i>	3 (3+0)
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Prasyarat : Tiada

Kursus ini merangkumi sejarah perkembangan, teori, konsep dan akta serta bentuk koperasi pertanian di Malaysia serta membincangkan fungsi, peranan, strategi pengurusan dan kepentingannya kepada industri pertanian. Persamaan dan perbezaan antara organisasi pertubuhan pertanian dinegara lain turut ditekankan.
This course encompasses the historical development, theories, concept, Acts and types of the agricultural cooperatives in Malaysia including discussion on function, role, management strategies and their importance to the agriculture industry. The similarities and differences vis-a-vis other country's agricultural organizations are also emphasized.

PPT4209	Sistem Pengurusan Perladangan / <i>Plantation Management Systems</i>	3 (3+0)
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Prasyarat : Tiada

Kursus ini merangkumi prinsip dan ciri pengurusan perladangan dan cara membuat keputusan dalam organisasi dan operasi sektor perladangan. Pelbagai akta yang berkaitan dengan pengurusan perladangan, isu dan cabaran terkini dalam sektor perladangan juga akan dibincangkan.
This course covers the principles and characteristics of plantation management and decision-making methods in the organization and operation of the plantation sector. Various acts related to plantation management, current issues and challenges in the plantation sector will also be discussed.

PPT4210	Pembangunan Sumber Manusia Dalam Sektor Pertanian / <i>Human Resource Development in Agricultural Sector</i>	3 (3+0)
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Prasyarat : Tiada

Kursus ini merangkumi teori pembangunan sumber manusia, asas keperluan kerja, program latihan, proses penilaian, dan pengurusan sumber manusia dalam organisasi sektor pertanian.
This course covers the theory of human resource development, basic job requirement, training program, evaluation process, and management of human resource in agricultural organizations.

PPT4211	Pengurusan Strategik Perniagaantani / <i>Agribusiness Strategic Management</i>	3 (3+0)
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Prasyarat : PPT3201

Kursus ini merangkumi teori dan penilaian dalam pengurusan strategik untuk organisasi asas tani bagi persediaan menghadapi perubahan dalam persekitaran masa depan secara sistematik dan strategik. Pengurusan rantai penawaran, pilihan strategi serta kaedah pengurusan strategik turut dibincangkan.
This course encompasses theory and assessment in strategic management for agri-based organization to prepare systematically and strategically for changes in future environment. Supply chain management, strategy options, and methods of strategic management are also discussed.

PPT4212	Perancangan Dan Pengurusan Projek Perniagaantani / <i>Agribusiness Project Planning and Management</i>	3 (3+0)
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Prasyarat : Tiada

Kursus ini merangkumi teori dan teknik dalam merancang dan menilai projek perniagaantani. Pelbagai teknik rangkaian pengurusan perniagaantani untuk menganalisis dan membandingkan daya maju projek perniagaantani yang berbeza juga diberi penekanan.
This course encompasses the theory and techniques in planning and evaluating agribusiness project. Various agribusiness management network techniques to analyze and to compare the viability of different agribusiness projects are also emphasized.

PPT4305	Pemasaran Perniagaantani / <i>Agribusiness Marketing</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi konsep pemasaran firma dan sistem perniagaantani, penilaian prestasi sistem pemasaran dan memahami persekitaran pemasaran. Pembentukan pelan pemasaran dan pengurusan campuran pemasaran turut dibincangkan. <i>This course encompasses the marketing concept in the agribusiness firm and system, evaluation of performance of marketing systems and understanding of marketing environment. The development of a marketing plan and management of the marketing mix are also discussed.</i>		
PPT4306	Perniagaantani Digital / <i>Digital Agribusiness</i>	3 (2+1)
Prasyarat : PPT4305 Kursus ini merangkumi pemahaman tentang perniagaan digital dalam sektor perniagaantani. Perkembangan, kepentingan, dan strategi dalam perniagaantani digital serta pengurusan perniagaantani dalam era data raya diberi penekanan. Kaedah membangunkan laman sesawang perniagaan, analisis strategi pemasaran serta undang-undang siber juga dibincangkan. <i>This course covers an understanding of digital business in the agribusiness sector. The development, importance and strategies in the digital business as well as agribusiness management in the era of big data are emphasized. Methods in developing business websites, analysis of marketing strategies as well as cyber law are also discussed</i>		
PPT4307	Pengurusan Rantaian Bekalan Produk Makanan / <i>Food Product Supply Chain Management</i>	3 (3+0)
Prasyarat : Tiada Kursus ini meliputi konsep pengurusan rantaian bekalan dalam pengeluaran, pemprosesan dan pemasaran produk makanan. Isu dan cabaran pengurusan rantaian bekalan makanan serta strategi yang bersesuaian untuk menambah baik pengurusan rantaian bekalan produk makanan juga diberi penekanan. <i>This course covers the concept of supply chain management in production, processing and marketing of food products. Issues and challenges of food supply chain management as well as appropriate strategies to improve food product supply chain management are also emphasized.</i>		
PPT4308	Pemasaran Perniagaantani Antarabangsa / <i>International Agribusiness Marketing</i>	3 (3+0)
Prasyarat : PPT4305 Kursus ini merangkumi pengenalan kepada pemasaran antarabangsa termasuk analisis persekitaran yang meliputi dimensi ekonomi, budaya, politik dan peraturan antarabangsa. Pengurusan dan pengawal-seliaan pemasaran antarabangsa juga ditekankan. <i>This course encompasses an overview of international marketing including its environment, economic, cultural, political and legal dimensions. The management and coordination of international marketing are also emphasized.</i>		
PPT4351	Pengurusan Perniagaantani Kecil Dan Sederhana / <i>Small and Medium Agribusiness Management</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi konsep dan strategi pengurusan perniagaantani kecil dan sederhana yang berdaya saing. Kaedah penilaian daya maju perniagaan, penerokaan modal, teknik mitigasi risiko dan strategi pertumbuhan perniagaan juga akan dibincangkan. <i>This course encompasses the management concepts strategies of competitive small and medium agribusiness enterprises. Methods to evaluate the feasibility of business, capital procurement, risk mitigation techniques, and business growth strategies will also be discussed.</i>		

PPT4402	Pengurusan Kewangan Perniagaantani / <i>Agribusiness Financial Management</i>	3 (3+0)
Prasyarat : Tiada Kursus ini meliputi penggunaan prinsip pengurusan kewangan untuk mencapai penggunaan sumber modal yang optimum dalam operasi perniagaantani. Aspek pengurusan kredit yang meliputi pembiayaan secara pinjaman, pajakan dan sewa beli bagi menganalisis masalah pengurusan kewangan perniagaantani juga dibincangkan. <i>This course covers the application of financial management principles in achieving optimum use of capital resources for agribusiness operations. Aspects of credit management which include financing through loans, leases and hire purchase to analyze the financial management problems of agribusinesses are also discussed.</i>		
PPT4501	Polisi dan Peraturan Perniagaantani / <i>Agribusiness Policy And Regulations</i>	3 (3+0)
Prasyarat : EPT3104 Kursus ini merangkumi teori ekonomi dan pelbagai polisi pertanian negara yang terkini. Aplikasi teori ekonomi bagi menilai kesan dan cabaran polisi dan peraturan dalam industri perniagaantani berdasarkan isu persekitaran dan strategik penyelesaian masalah juga diberi penekanan. <i>This course covers economic theory and various current national agricultural policies. The application of economic theory to assess the impact and challenges of policies and regulations in the agribusiness industry based on environmental issues and problem solving strategies is also emphasized.</i>		
PPT4503	Isu Semasa Perniagaantani / <i>Current Issues in Agribusiness</i>	3 (3+0)
Prasyarat : Tiada Kursus ini meliputi isu semasa dalam sektor perniagaantani yang merangkumi pelbagai aspek di peringkat dalam dan luar negara. Kaitan di antara industri asas tani dalam kelestarian alam sekitar juga dibincangkan. Pengalaman dan perkembangan semasa dalam amalan di industri asas tani juga ditekankan. <i>This course covers current issues in agribusiness sector including various aspects in local and international level. Relationships between agri-based industry with environment sustainability is also discussed. Experience and current development in agri-based industry practices are also emphasized.</i>		
PPT4602	Kaedah Penyelidikan Perniagaantani / <i>Agribusiness Research Methods</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi konsep dan kepentingan penyelidikan di dalam bidang perniagaantani. Proses menjalankan penyelidikan saintifik, skala dan pengukuran serta kaedah statistik yang bersesuaian untuk menganalisis data, penyediaan laporan dan memberi cadangan dalam penyelidikan perniagaantani turut ditekankan. <i>This course covers the concept and importance of research in agribusiness. The process of conducting scientific research, scales and measurements as well as appropriate statistical methods for analyzing data, preparing reports and making recommendations in agribusiness research are also emphasized.</i>		
PPT4702	Keusahawanan Perniagaantani / <i>Agribusiness Entrepreneurship</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep keusahawanan, ciri usahawan, dan pembinaan pelan perniagaan. Sifat kepemimpinan dan kemahiran berfikir secara kritis ditekankan dalam pelaksanaan projek, strategi pemasaran, dan isu semasa keusahawanan. <i>This course covers the concepts of entrepreneurship, the characteristics of an entrepreneur, and construction of a business plan. Leadership traits and critical thinking skills are emphasized for implementation of project, marketing strategies and current issues of agri-entrepreneurship.</i>		

PPT4801	Teknik Kuantitatif Perniagaantani / <i>Quantitative Techniques In Agribusiness</i>	3 (2 + 1)
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Prasyarat : PPT3801

Kursus ini meliputi konsep dan teori kaedah kuantitatif untuk menganalisis masalah perniagaantani. Aplikasi teknik kuantitatif dan proses membina model kuantitatif untuk menilai masalah dalam pengurusan perniagaantani dan ekonomi pertanian serta penyediaan laporan dan cadangan untuk penyelesaian masalah dalam pengurusan perniagaantani dan ekonomi pertanian juga diberi penekanan.

This course covers the concepts and theories of quantitative methods for analyzing agribusiness problems. The application of quantitative techniques and the process of building quantitative models to assess problems in agribusiness management and agricultural economics as well as the preparation of reports and recommendations for problem solving in agribusiness management and agricultural economics are also emphasized.

PPT4901	Latihan Industri Perniagaantani / <i>Agribusiness Industrial Training</i>	12 (0 + 12)
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Prasyarat : Tiada

Kursus ini merangkumi penempatan pelajar di syarikat berasaskan perniagaantani selama satu semester untuk mengikuti latihan dalam semua aspek pengurusan. Semasa menjalani latihan ini pelajar dapat menimba pengetahuan mengenai pengurusan pelbagai isu perniagaantani dari agensi awam atau syarikat swasta.

This course encompasses placement of students in agribusiness firms for one semester to undergo training in all aspects of management. During the training period, students will gain knowledge and practical experience on the management of various agribusiness issues from public agency or private companies.

PPT4959	Disertasi Bachelo / <i>Bachelor Dissertation</i>	6 (0 + 6)
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Prasyarat : Tiada

Kursus ini merangkumi pengetahuan dan pengalaman praktik dalam mengendalikan dan melaporkan sesuatu projek penyelidikan. Topik akan dikenalpasti dalam bidang pengkhususan untuk dijalankan penyelidikan dengan bimbingan seorang penyelia. Proses merancang, melaksana dan melaporkan hasil kajiannya turut dibincangkan.

This course encompasses knowledge and practical experience in conducting and reporting of a research project. A topic will be identified in a specialization area for a research will be conducted with the guidance of a supervisor. The process of plan, implementing and reporting the findings of the research are also discussed.

EPT3104	Ekonomi Pertanian / <i>Agricultural Economics</i>
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Prasyarat : Tiada

Kursus ini merangkumi aplikasi konsep asas teori ekonomi. Aspek aplikasi teori ekonomi dalam bidang pertanian serta kaitannya dengan struktur pasaran, isu penawaran dan permintaan, dasar kerajaan, dan perdagangan antarabangsa juga dibincangkan.

This course covers the basic concept of economic theories. Aspects of the application of economic theory in agriculture as well as its relation to market structure, supply and demand issues, government policy, and international trade are also discussed.

EPT3105 Ekonomi Pertanian/
Agricultural Economics

Prasyarat : Tiada

Kursus ini memperkenalkan konsep asas ekonomi dan aplikasinya dalam sektor pertanian, termasuk faktor yang mempengaruhi pasaran serta dasar yang berkaitan. Kaedah menganalisis isu ekonomi pertanian dan membuat keputusan berasaskan teori ekonomi turut dibincangkan.

This course introduces basic economic concepts and their application in the agricultural sector, including factors influencing the market and relevant policies. Methods for analyzing agricultural economic issues and making decisions based on economic theory are also discussed

EPT3203 Pengurusan Ladang/
Farm Management

3 (3+0)

Prasyarat : Tiada

Kursus ini meliputi penggunaan teori dan prinsip ekonomi di dalam pengurusan ladang. Pelbagai pendekatan dan kaedah yang digunakan dalam pengurusan dan analisis prestasi sesebuah perniagaan ladang dan pengurusan risiko juga dibincangkan.

This course covers applications of economic theory and principles in farm management. Various approaches and methods used in the management and performance analysis of a farm business and risk management are also discussed.

EPT4201 Analisis Keputusan Pertanian/
Agricultural Decision Analysis

3 (3+0)

Prasyarat : PPT3801

Kursus ini meliputi konsep dan penggunaan model dalam proses pembuatan keputusan pertanian. Penstrukturan masalah dengan mengambilkira elemen masalah keputusan dan penggunaan teknik kuantitatif dan statistik yang bersesuaian untuk mengkaji risiko serta masalah pertanian kompleks turut ditekankan.

This course covers the concept and use of models in the agricultural decision-making process. Problem structuring by taking into account the elements of a decision problem and by using appropriate quantitative and statistical techniques to study complex agricultural risks and problems are also emphasized.

EPT4202 Pengurusan Risiko Pertanian/
Agricultural Risk Management

3 (3+0)

Prasyarat : EPT3104

Kursus ini merangkumi konsep dan prinsip pengurusan risiko bagi pengeluaran dan pemasaran pertanian. Teori pengurusan risiko dan pelbagai teknik analisis berdasarkan jenis risiko yang mungkin berlaku di pasaran pertanian untuk membuat keputusan yang lebih tepat juga diberi penekanan.

This course covers the concepts and principles of risk management for agricultural production and marketing. Risk management theory and various analysis techniques based on the types of risks that may occur in the agricultural market in making more accurate decisions are also emphasized.

EPT4501 Ekonomi dan Dasar Pembangunan Pertanian/
Economics of Agricultural Development and Policy

3 (3+0)

Prasyarat : Tiada

Kursus ini merangkumi pelbagai teori pembangunan, polisi dan isu semasa pertanian di negara maju, negara membangun dan negara kurang membangun. Ciri dan implikasi polisi untuk pembangunan ekonomi mengikut tahap kemajuan negara dan strategi untuk pembangunan dan pertumbuhan ekonomi pertanian negara diberi penekanan.

This course encompasses various agricultural development theories, policies and current issues in developed, developing and less developed countries. The characteristics and implications of policies for economic development according to the level of national development and strategies for the development and growth of the national agricultural economy are emphasized.

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Pengurusan Tanah / Department of Land Management

SST3005	Sains Tanah Asas / <i>Fundamentals Of Soil Science</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi pembentukan tanah, sifat kimia, fizik dan biologi tanah, pemakanan tumbuhan dan baja, pengkelasan, survei dan penilaian tanah. <i>This course encompasses soil formation, physical, chemical and biological properties, plant nutrition and fertilizers, soil classification, survey and evaluation.</i>		
SST3100	Pengurusan Kesuburan Dan Kelestarian Tanah / <i>Soil Fertility Management and Soil Sustainability</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi kepentingan sifat fizik, kimia dan biologi tanah terhadap produktiviti pertanian. Amalan pengurusan nutrien tanaman, sisa organik, air tanah dan tanah terdegradasi secara lestari turut dibincangkan. <i>This course covers the importance of the physical, chemical and biological properties of soil to agricultural productivity. Sustainable management practices of crop nutrients, organic waste, groundwater and degraded soil are also discussed.</i>		
SST3201	Tanah-Tanah Di Malaysia / <i>Soils Of Malaysia</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi prinsip survei tanah, pemerihalan profil tanah, pengkelasan tanah di Malaysia mengikut siri dan pengurusan am kumpulan tanah utama. <i>This course encompasses the principles of soil survey, description of soil profile, classification of soils in Malaysia according to series and the general management of major soil group.</i>		
SST3502	Ekologi Tanah / <i>Soil Ecology</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi prinsip ekologi tanah, interaksi antara organisma tanah dan persekitarannya, peranan tanah dalam ekosistem, proses biologi dan faktor mempengaruhi aktiviti organisma, kesan pencemaran terhadap organisma tanah, kawalan biologi dan bioremediasi tanah tercemar. <i>This course covers the principles of soil ecology, interactions between soil organisms and their environments, role of soils in ecosystem, biological processes and factors influencing the activities of soil organisms, the impact of pollution on soil organisms, biological control and bioremediation of polluted soils.</i>		

SST3602	Pengurusan Tanah Lestari / <i>Sustainable Soil Management</i>	3 (2+1)
Prasyarat : SST3005 atau setara Kursus ini merangkumi konsep pertanian secara lestari, cara menilai kesuburan tanah dan keperluan baja, peningkatan dan penyelenggaraan kualiti tanah, kesan dan kawalan degradasi tanah, tanah sebagai sinki untuk bahan cemar dan cara mengurusnya. <i>This course encompasses the concept of sustainable agriculture, methods of assessing soil fertility and fertilizer requirement, improvement and maintenance of soil quality, impacts and control of soil degradation, soil as a sink for pollutants and its management.</i>		
SST3603	Pengurusan Kesuburan Tanah / <i>Soil Fertility Management</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi prinsip, kepentingan dan amalan pengurusan kesuburan tanah, nutrien dan air untuk penghasilan tanaman. Kesan amalan ladang terhadap kesuburan tanah untuk pengeluaran tanaman dibincangkan. <i>This course covers the principles, importance and practices in managing soil fertility, nutrient and water for crop production. Effects of field management on soil fertility for crop production are discussed.</i>		
SST3604	Kesuburan Tanah Tanaman Perladangan / <i>Soil Fertility for Plantation Crops</i>	3 (1+2)
Prasyarat : Tiada Kursus ini meliputi prinsip, kepentingan dan amalan pengurusan kesuburan tanah, nutrien dan air untuk penghasilan tanaman perladangan, kesan amalan ladang terhadap kesuburan tanah dalam mengoptimumkan pengeluaran tanaman perladangan dengan menggunakan Amalan Pertanian Baik (GAP). <i>This course covers the principles, importance and practices in managing soil fertility, nutrient and water for plantation crop production, effects of field management on soil fertility for optimum plantation crop production using Good Agricultural Practices (GAP).</i>		
SST3802	Pengurusan Sisa Pertanian / <i>Agricultural Waste Management</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi pengenalan kepada jenis sisa pertanian, masalah pencemaran alam sekitar dan pengurusan secara mesra alam untuk pelupusan dan/atau kitaran semula, dan aspek peraturan alam sekitar. <i>This course covers introduction to the different types of agricultural wastes, environmental problems and environmentally friendly waste management for disposal and/or recycling, and environmental regulatory aspects.</i>		
SST4005	Kaedah Penyelidikan Sains Tanah / Research Methods in Soil Science	3 (2+1)
Prasyarat : SST3005 Kursus ini meliputi kaedah dan instrumentasi yang biasa digunakan dalam bidang mikrobiologi, fizik dan kimia tanah. Kesesuaian kaedah dan instrumen dengan topik penyelidikan di dalam bidang sains tanah turut dibincangkan. <i>This course encompasses common methods and instruments used in soil science especially in the fields of soil microbiology, physics and chemistry. The suitability of methodology and instruments with research topics in the field of soil science will be discussed.</i>		

SST4201	Pembentukan dan Pengelasan Tanah / <i>Soil Formation and Classification</i>	3 (2+1)
Prasyarat : SST3005 atau setara		
Kursus ini meliputi morfologi tanah, luluhawa dan pembentukan tanah, indeks kestabilan, sintesis mineral dalam tanah dan prinsip pengelasan tanah mengikut sistem pengelasan tanah yang berlainan. <i>This course covers soil morphology, weathering and soil formation, stability indices, mineral synthesis in soil and the principles of soil classification according to different soil classification systems.</i>		
SST4202	Survei Tanah Dan Pengelasan Kesesuaian Guna Tanah / <i>Soil Survey and Land Use Suitability Classification</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini meliputi teknik survei tanah, <i>Legend Map</i> tanah dan pengelasan kesesuaian guna tanah untuk pertanian. Pelajar didedahkan kepada pelaksanaan survei tanah di agensi perladangan. <i>This course covers techniques of soil survey, soil Legend Map and land use suitability classification for agriculture. Students are exposed to soil survey activity in plantation agencies.</i>		
SST4302	Fizik Tanah / <i>Soil Physics</i>	3 (2+1)
Prasyarat : SST3005 atau setara		
Kursus ini merangkumi ciri fizik dan proses fizikal tanah seperti air tanah, hakisan tanah dan struktur tanah. <i>This course encompasses the soil physical characteristics and processes such as soil water, soil erosion, and soil structure.</i>		
SST4303	Pemuliharaan Tanah / <i>Soil Conservation</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi proses, faktor, kaedah pengawalan, ramalan hakisan dan kepentingan pemuliharaan tanah tropika. Kesan pemuliharaan tanah ke atas tanaman dan persekitaran dibincangkan. <i>This course encompasses the processes, factors, control measures, prediction of soil erosion and importance of tropical soil conservation. The impact of soil conservation on crop and the environment are discussed.</i>		
SST4304	Perubahan Iklim Dan Pertanian / <i>Climatic Change And Agriculture</i>	3 (2+1)
Prasyarat : SST3005		
Kursus ini merangkumi sains perubahan iklim, penyebab ianya berlaku serta kesannya terhadap pertanian pengeluaran makanan. Cara mitigasi dan permodelan perubahan iklim dibincang. <i>This course encompasses the science of climatic change, its causes as well as its effects on agriculture food production. Mitigation efforts and modeling of climatic change are discussed.</i>		
SST4402	Kimia Tanah / <i>Soil Chemistry</i>	3 (2+1)
Prasyarat : SST3005 atau setara		
Kursus ini meliputi proses utama kimia dalam persekitaran tanah termasuk mineralogi lempung silikat dan proses luluhawa, kimia koloid, keasidan dan kemasinan tanah, dan tindakbalas redoks dalam tanah. <i>This course covers the main chemical processes in the soil environment which include mineralogy of silicate clays and weathering process, colloidal chemistry soil acidity and salinity, and redox reactions in soil.</i>		

SST4501	Mikrobiologi Tanah Lanjutan / <i>Advanced Soil Microbiology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi ekologi tanah, faktor yang mempengaruhi aktiviti mikroorganisma tanah, kaedah pengiraan populasi dan pengasingan mikrob, interaksi tanah-tumbuhan-mikrob, transformasi mineral, kesan ekologi racun perosak, kawalan biologi, dan bioremediasi tanah.

This course covers soil ecology, soil factors affecting activities of soil microorganisms, methods of microbial population count and isolation, soil-plant-microbe interactions, mineral transformations, ecological effects of pesticides, biological control, and soil bioremediation.

SST4601	Kesuburan Tanah Lanjutan / <i>Advanced Soil Fertility</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi kitar biokimia unsur nutrien, mekanisme pengangkutan nutrien dalam fasa pepejal dan cecair, pengambilan nutrien daripada tanah, kekurangan dan keracunan nutrien, penilaian kesuburan tanah, pembajaan terutama tanah bermasalah, tafsiran data analitik dan syor praktikal berkaitan dengan pengendalian dan pembaikan kesuburan tanah.

This course covers biochemical cycling of nutrient elements, mechanisms of nutrient transport in solid and liquid phases, uptake of nutrients from soil, nutrient deficiencies and toxicities, evaluation of soil fertility, fertilization especially of problem soils, interpretation of analytical data and practical recommendation regarding maintenance and improvement of soil fertility.

SST4602	Kelestarian Perladangan Dan Pengurusan Kualiti / <i>Plantation Sustainability and Quality Management</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini menumpukan kepada amalan pertanian baik merangkumi amalan pemuliharaan tanah, air dan biodiversiti serta kualiti udara bagi mengurangkan impak kepada alam sekitar. Pengurusan kualiti merangkumi peningkatan sosial dan polisi kelestarian juga diberi penekanan.

This course focuses on good agriculture practices emphasizing on soil, water and biodiversity conservation and air quality to reduce environmental impact. Quality management which includes social enhancement and sustainability policies are emphasized

SST4701	Mineralogi Tanah / <i>Soil Mineralogy</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi jenis, pembentukan dan sifat mineral tanah serta pelbagai kaedah mengenalpasti mineral.

This course covers the types, formation and properties of soil minerals and various methods to identify minerals.

SST4801	Sains Tanah Persekitaran / <i>Environmental Soil Science</i>	3 (2+1)
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Prasyarat : SST3005 atau setara

Kursus ini merangkumi proses kontaminasi dan pencemaran tanah dari aspek fizikal, biologi dan kimia. Kaedah bagi mengawal dan mengatasi masalah kontaminasi dan pencemaran tersebut dibincangkan. Kaedah pengurusan tanah yang mesra alam dan lestari dalam bidang pertanian akan ditekankan.

This course encompasses the processes in soil contamination and pollution from the physical, biological and chemical aspects. Methods to control and to overcome soil contamination and pollution are discussed. Soil management practices which are sustainable and eco-friendly in agriculture are emphasized.

PRT4959	Disertasi Bachelor / <i>Bachelor Dissertation</i>	6 (0+6)
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Prasyarat : Tiada

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui rekabentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.

This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.

SPP4905	Pengurusan dan Pengoperasian Ladang / <i>Plantation Management and Operation</i>	14 (0+14)
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Prasyarat : PRT4959

Kursus ini merangkumi aspek komunikasi dalam organisasi korporat, pengurusan perosak, pengurusan penyakit dan rumpai, penuaian dan pemprosesan hasil, pengurusan tanah dan pengurusan ladang. Kajian kes untuk menyelesaikan masalah terpilih tentang pengurusan dan pengoperasian ladang bermula dari mengenalpasti masalah, pengumpulan data dan cadangan penyelesaian dan penambahbaikan juga dijalankan.

This course encompasses the aspects of communication in corporate organisation, management of pests, management of diseases and weeds, harvesting and processing, soil management and plantation management. A case study on solving selected plantation management and operational problems starting from problem identification, data collection and suggestions for solutions and improvement are also conducted.

SINOPSIS KURSUS | COURSE SYNOPSIS

Pejabat Dekan /
Dean's Office

PRT2009	Pertanian Dan Kehidupan / <i>Agriculture and Life</i>	2 (1+1)
Prasyarat : Tiada Kursus ini merangkumi amalan pengeluaran hasil pertanian menggunakan pelbagai teknik yang bersesuaian. Pengurusan penghasilan serta kaitan aktiviti pertanian dengan kelestarian alam sekitar juga dibincangkan. <i>This course covers agricultural production practices using a variety of appropriate techniques. Production management and relationship between agricultural activities and environmental sustainability are also discussed.</i>		
PRT3003	Pertanian, Persekitaran dan Sistem Makanan / <i>Agriculture, Environment And Food System</i>	3 (3+0)
Prasyarat : Tiada Kursus ini meliputi aktiviti pertanian yang mempengaruhi kualiti alam sekitar dan pengeluaran produk komoditi pertanian yang baik melalui sistem pengurusan persekitaran untuk mengekalkan kualiti tanah, air dan udara bersih. Isu semasa tentang kepelbagaian produk, kualiti makanan, keselamatan dan kesihatan makanan, etika serta cabaran dalam era globalisasi dibincangkan. <i>This course covers agricultural activities influencing environmental quality and production of good agricultural product commodities through environmental management system to maintain soil and water quality, and clean air. Environmental issues on product diversity, food quality, food security and safety, ethics and challenges in the era of globalization are discussed.</i>		
PRT3005	Sistem Pengeluaran Tanaman / <i>Crop Production System</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi jenis tanaman perladangan, makanan, hortikultur, tumbuhan herba dan ubatan, faktor yang mempengaruhi penentuan jenis dan lokasi dalam sistem pengeluaranannya, penanaman dan teknik pengurusan tanaman. <i>This course encompasses different crop types such as plantation, food and horticulture, herbal and medicinal plants, factors that influence the choice of types and locations in the production systems, planting and crop management techniques.</i>		

PRT3006	Prinsip Pertanian Lestari / <i>Principles Of Sustainable Agriculture</i>	2 (2+0)
Prasyarat : Tiada Kursus ini merangkumi keperluan dan perkembangan pertanian lestari. Amalan pertanian yang mengambil kira elemen pembangunan sosial, pertumbuhan ekonomi dan pemuliharaan alam sekitar dibincangkan. <i>The course encompasses the need and development of sustainable agriculture. Agricultural practices that consider the elements of social development, economic growth and environmental conservation are discussed.</i>		
PRT3007	Sistem Pengeluaran Pertanian / <i>Agriculture Production System</i>	3 (3+0)
Prasyarat : Tiada Kursus ini memperkenalkan konsep pengeluaran pertanian lestari yang menghubungkan antara faktor pengeluaran, pengurusan dan aktiviti lepastuai. Sistem pengeluaran tanaman, penternakan dan akuakultur akan dibincangkan dengan contoh yang berkenaan. <i>This course introduces the concept of sustainable agricultural production system which relates interaction of the production factors, management and post-harvest activities. Systems for crop, livestock and aquaculture production will be discussed with relevant examples.</i>		
PRT3008	Keselamatan Dan Kesejahteraan Perladangan / <i>Plantation Safety and Security</i>	3 (1+2)
Prasyarat : Tiada Kursus ini merangkumi prinsip keselamatan dan kesejahteraan pekerja, penilaian risiko, perancangan keselamatan dan pengurusan insiden dalam industri perladangan. <i>This course covers principles of occupational safety and security, risk assessment, security planning and incident management in the plantation industry.</i>		
PRT3201	Asas Statistik Bagi Sains Pertanian / <i>Elementary Statistics For Agricultural Science</i>	2 (2+0)
Prasyarat : Tiada Kursus ini merangkumi kaedah statistik asas. Tajuk-tajuk termasuk statistik keperihalan, kebarangkalian, pensampelan rawak, ujian hipotesis, jangkaan, regresi linear dan korelasi. <i>The course encompasses basic statistical methods. Topics include descriptive statistics, probability, random sampling, tests of hypothesis, estimation, simple linear regression and correlation.</i>		
PRT3202	Rekabentuk Dan Analisis Eksperimen / <i>Experimental Design And Analysis</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi kaedah melaksanakan eksperimen yang merangkumi rekabentuk eksperimen, pensampelan, analisis dan interpretasi data dan membuat rumusan. <i>This course covers the methods of conducting experiments including experimental design, sampling, data analysis and interpretation, and formulating conclusions.</i>		
PRT3204	Biometri / <i>Biometri</i>	4 (3+1)
Prasyarat : Tiada Kursus ini merangkumi kaedah statistik untuk merekabentuk eksperimen, menganalisis data dan membuat kesimpulan daripada keputusan eksperimen <i>This course encompasses statistical methods to design an experiment, to analyze the data and to make conclusions from the experimental results</i>		

PRT3205	Statistik Gunaan / <i>Applied Statistics</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi prinsip analisis statistik, pembinaan rekabentuk eksperimen, dan analisis data statistik. Pelaporan kesimpulan hasil keputusan eksperimen turut dibincangkan. <i>This course covers the hypothesis testing for field study, data analysis using open-source statistical software and concluding the experimental results based on the hypothesis tested for the field study condition.</i>		
PRT3401	Kimia Pertanian / <i>Agricultural Chemistry</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi prinsip asas kimia yang penting dalam bidang pertanian termasuk proses kimia dalam pertanian, prinsip analisis bahan pertanian dan pengenalan kepada bahan kimia organik asli dan sintetik dalam pertanian. <i>This course encompasses the important basic chemical principles in agriculture which includes chemical processes in agriculture, analytical principles of agricultural materials, and introduction to natural and synthetic organic chemicals in agriculture.</i>		
PRT3402	Biokimia Pertanian / <i>Agricultural Biochemistry</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi proses biokimia dalam sistem eukariot, struktur dan peranan biomolekul, pemangkinan dan kawalan tindakbalas biokimia. <i>This course encompasses biochemical processes in eukaryotic system, structure and roles of biomolecules, catalysis and control of biochemical reactions</i>		
PRT3403	Kimia Dan Biokimia Pertanian / <i>Agricultural Chemistry and Biochemistry</i>	4 (3+1)
Prasyarat : Tiada Kursus ini merangkumi prinsip kimia dan biokimia yang utama dalam bidang pertanian. Struktur dan peranan biomolekul, enzim dan hormon serta pengurusan maklumat kimia dan biokimia dalam bidang pertanian ditekankan. <i>This course encompasses the main chemical and biochemical principles in agriculture. The structure and roles of biomolecules, enzymes and hormones, and management of chemical and biochemical information in agriculture are emphasized.</i>		
PRT3501	Mikrobiologi Pertanian / <i>Agricultural Microbiology</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi prinsip mikrobiologi serta aplikasi dan peranan mikrob di dalam alam sekitar dan industri. <i>This course covers principles of microbiology as well as the application and role of microbes in the environment and industry.</i>		
PRT4002	Pertanian Presisi / <i>Precision Agriculture</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep dan amalan lestari dalam pertanian presisi yang lestari. Pengukuran variabiliti menerusi berbagai prosedur pensampelan, pemetaan nutrien, hasil dan perosak, interpolasi corak ruang, sistem maklumat geografi, permodelan dan simulasi pertumbuhan tanaman, dan penggunaan data penderiaan jarak jauh. <i>This course encompasses concepts and sustainable practices in precision agriculture. Variability measurements using various sampling procedures, nutrient, yield and pest mapping, interpolation of spatial patterns, geographical information system, modelling and simulation of crop growth, and use of remote sensing data.</i>		

PRT4041	Sistem Pengeluaran Organik / <i>Organic Production Systems</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi prinsip dan amalan dalam pengeluaran makanan organik dengan menitikberatkan sistem perladangan lestari, sistem tanaman, kawalan perosak dan penyakit, piawaian organik dan pensijilan, sosioekonomi dan pemasaran. <i>This course covers the principles and practices of organic food production with emphasis on sustainable farming systems, pest and disease control, organic standard and certification, socioeconomics and marketing.</i>		
PRT4201	Penilaian Sumber Pertanian / <i>Agricultural Resource Evaluation</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi penilaian sumber pertanian termasuk sumber tanah, air, genetik, tenaga dan sumber manusia, dan ramalan hasil pertanian. <i>This course encompasses evaluation of various resources for agriculture which include soil, water, genetic, energy and human resources, and yield forecasting.</i>		
PRT4301	Permodelan Dan Simulasi Komputer Dalam Pertanian / <i>Modelling And Computer Simulation In Agriculture</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi pemerihalan proses biofizikal dan proses mikrometeorologi tanaman dalam bentuk persamaan matematik. <i>This course covers the description of biophysical and crop micrometeorological processes in the form of mathematical equations.</i>		
PRT4302	Agroiklim / <i>Agroclimatology</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi komponen cuaca, kaedah pengukuran dan kesan perubahan iklim terhadap produktiviti pertanian. Impak kegiatan manusia terhadap perubahan iklim serta cadangan beberapa amalan pertanian untuk menangani impak perubahan iklim dibincangkan. <i>This course covers weather components, measurement methods and the effects of climate change on agriculture productivity. Impacts of human activities on climate change and suggestions of several agriculture practices to mitigate the impacts of climate change are discussed.</i>		
PRT4901	Projek Ilmiah Tahun Akhir / <i>Final Year Academic Project</i>	12 (0+12)
Prasyarat : Tiada Melalui kursus ini, pelajar didedahkan kepada persekitaran kerja yang sebenar di industri/organisasi. Latihan merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi ditekankan. <i>In this course, students are exposed to real working environment in industries/organizations. Training includes application of the theoretical and practical aspects that have been studied with current practices in the workplace. Problem solving and communication skills are emphasized.</i>		
PRT4999	Projek Ilmiah Tahun Akhir / <i>Final Year Academic Project</i>	8 (0+8)
Prasyarat : Tiada Kursus ini merangkumi perancangan, pelaksanaan, analisis dan penterjemahan data, menyediakan dan membentangkan laporan saintifik. <i>This course encompasses planning, implementation, data analysis and interpretation, preparation and presentation of scientific report.</i>		

PRT4959	Disertasi Bachelo/ <i>Bachelor Dissertation</i>	6 (0+6)
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Prasyarat : Tiada

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui rekabentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.

This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.

SINOPSIS KURSUS | COURSE SYNOPSIS

Jabatan Teknologi Pertanian Department of Agriculture Technology

TKP3001	Teknologi Pertanian Modern / <i>Modern Agricultural Technology</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi penggunaan teknologi pertanian moden didalam mempertingkatkan hasil dan kualiti produk pertanian. Aspek sosial dalam konteks teknologi pertanian moden turut dibincangkan. <i>This course encompasses the use of agricultural technology to increase yield and quality of agricultural produce. Social aspects in the context of agricultural technology are discussed.</i>		
TKP3002	Amalan Ladang: Akuakultur / <i>Farm Practices : Aquaculture</i>	1 (0+1)
Prasyarat : Tiada		
Kursus ini merangkumi amalan penyeliaan kolam, pengendalian unit penetasan, pembiakan, larvikultur, pengurusan kolam dan tangki, pengeluaran, pemakanan dan pemanenan ikan dan udang. <i>This course covers practices in pond preparation managing hatchery unit, breeding, larviculture, pond and tank management, production, feeding and harvesting of fish and prawn.</i>		
TKP3104	Bioteknologi Pertanian / <i>Agricultural Biotechnology</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi bioteknologi pertanian, teknik dan aplikasinya pada tumbuhan dan haiwan. Teknologi terkini dalam bioteknologi dibincangkan. <i>This course encompasses agricultural biotechnology, its technique and applications on plants and animals. Current technology in biotechnology will be discussed.</i>		
TKP3106	Pembiakbakaan Tanaman Dan Pengeluaran Biji Benih / <i>Crop Breeding and Seed Production</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi teknik penghasilan dan pengurusan varieti baharu serta pengekalan ketulenan baka. Perkembangan, kualiti, ujian, teknik pengeluaran, pemprosesan dan penyimpanan biji benih turut dibincangkan. <i>This course encompasses techniques for the production and management of new varieties as well as the maintenance of breed authenticity. Development, quality, testing, production techniques, processing and storage of seeds are also discussed.</i>		
TKP3201	Pengembangan Pertanian / <i>Agricultural Extension</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi sejarah, kaedah, perencanaan program dan pendedahan kepada situasi semasa kaedah pengembangan pertanian. <i>This course covers the history, methods, program planning and exposure to the current situation of agricultural development methods.</i>		

TKP3205	Komunikasi Pengembangan Dan Demonstrasi Kaedah Pertanian / <i>Agriculture Extension Communication and Method Demonstration</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi hubungkait pembangunan model komunikasi pengembangan dengan demonstrasi kaedah. Pelaksanaan demonstrasi kaedah dan keberkesanan aktiviti pengembangan pertanian dan kerja lapangan dibincangkan. <i>This course encompasses relationship between development of extension communication model and method demonstration. Implementation of method demonstration and effectiveness of agriculture extension activities and field work are discussed.</i>		
TKP3206	Pengembangan Akuakultur / <i>Aquaculture Extension</i>	2 (1+1)
Prasyarat : Tiada Kursus ini merangkumi kemahiran dan pengetahuan untuk membangunkan program dan latihan yang berkesan dalam khidmat pengembangan akuakultur kepada komuniti. Projek pengembangan dalam akuakultur dilaksanakan berdasarkan pembangunan ekonomi dan komuniti <i>This course covers the skills and knowledge to develop effective programs and training in aquaculture extension services to community. Extension projects in aquaculture are conducted based on economics and community development</i>		
TKP3301	Taman Rekreasi Pertanian / <i>Agriculture Recreational Park</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi konsep, pembangunan, khidmat sokongan serta perancangan aktiviti taman rekreasi pertanian untuk penambah nilai ladang pertanian. Proses pembangunan taman rekreasi pertanian dari sebuah ladang yang sedia ada serta menggunakan kemudahan di ladang untuk perancangan aktiviti rekreasi pertanian. Kepentingan penyelenggaraan bagi kepuasan pengunjung juga akan di bincangkan. <i>This course covers the concept, development, support services also recreational activities of agriculture park to adding value to farm. The process of developing agriculture recreational park from an established farm and use the available facilities to plan agriculture recreational activities. The importance of maintenance for visitor satisfaction will also be discussed.</i>		
TKP3401	Teknologi Maklumat Dan Komunikasi Untuk Pertanian / <i>Information and Communication Technology for Agriculture</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi ICT dan pemprosesan, pengaliran maklumat dan aplikasi ICT dalam pertanian. <i>This course encompasses ICT and processing, information flows and ICT applications in agriculture.</i>		
TKP3402	Perancangan dan Pelaksanaan Penderiaan Jarak Jauh / <i>Planning And Implementation of Remote Sensing Project</i>	4 (2+2)
Prasyarat : Tiada Kursus ini menekankan perancangan dan pelaksanaan projek yang melibatkan penggunaan penderia jarak jauh. Cadangan penyelesaian masalah pengurusan, pengeluaran pertanian dan integrasi maklumat berdasarkan pemilihan penderia, dan teknik perolehan imej turut dibincangkan. <i>This course emphasizes the planning and implementation of projects involving the use of remote sensors. Proposed problem-solving management, agricultural production and information integration based on sensor selection, and image acquisition techniques are also discussed.</i>		

TKP3403	Pengurusan Internet Benda (IoT) / <i>Internet of Things Management</i>	3 (3+0)
Prasyarat : Tiada Kursus ini merangkumi aplikasi IoT untuk memperbaiki amalan pertanian dan memacu ke arah pertanian digital. Aspek pemasangan dan penyelenggaraan sistem IoT diberi penekanan serta implikasi IoT terhadap rantai bekalan pertanian juga dibincangkan. <i>The course covers IoT applications to improve agricultural practices and drive towards digital agriculture. Aspects of installation and maintenance of IoT systems are given emphasis as well as the implications of IoT on the agricultural supply chain are also discussed.</i>		
TKP3405	Kebolehkesanan Makanan / <i>Food Traceability</i>	2 (2+0)
Prasyarat : Tiada Kursus ini merangkumi konsep, sistem, dan teknologi kebolehkesanan makanan dalam sektor pertanian serta penilaian bagaimana sistem kebolehkesanan makanan dilaksanakan dalam pelbagai industri makanan yang memfokuskan kepada pengeluaran makanan dan kualiti makanan. Kelebihan dan halangan aplikasi sistem kebolehkesanan makanan juga dibincangkan. <i>This course covers the concepts, systems, and technologies of food traceability in the agricultural sector as well as an assessment of how food traceability systems are implemented in various food industries focusing on food production and food quality. The advantages and disadvantages of the application of food traceability systems are also discussed.</i>		
TKP3501	Mekanisasi Pertanian Dan Pengairan / <i>Agricultural Mechanisation and Irrigation</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi rekabentuk, kesesuaian, pengendalian dan pengoperasian jentera utama yang digunakan dalam pengeluaran pertanian. <i>This course encompasses designs, suitability, handling and operation of major machineries used for agricultural production.</i>		
TKP3502	Pengoperasian Dron / <i>Drone Operation</i>	3 (2+1)
Prasyarat : Tiada Kursus ini merangkumi pengoperasian dron melibatkan pelbagai jenis dron dalam aplikasi pertanian. Pemilihan model, pengendalian penerbangan, pemilihan jenis penderia dan teknik penerbangan dron kepada komuniti teknologis turut dibincangkan. <i>This course covers drone operation involving various types of drones in agricultural applications. Model selection, flight handling, sensor type selection and drone flying techniques to the technological community are also discussed.</i>		
TKP3503	Pemprosesan Tanaman Perladangan / <i>Plantation Crop Processing</i>	3 (2+1)
Prasyarat : Tiada Kursus ini meliputi pengenalan kepada pemprosesan tanaman perladangan. Pengajaran akan memfokuskan kepada pemahaman terhadap operasi yang terlibat dalam pemprosesan tanaman perladangan. Pelajar juga akan didedahkan kepada pengenalan pastian dan pemilihan operasi pemprosesan yang sesuai untuk hasil pemprosesan yang optimum. <i>This course covers the introduction of plantation crop processing. The teaching will focus on the understanding of the operations involved in the processing of plantation crop. Students will also be exposed to the identification and selection of appropriate processing operations for optimum processing results</i>		

TKP3504	Pengurusan Automasi dan Robotik / <i>Automation and Robotics Management</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penyelesaian masalah melibatkan penggunaan sistem automasi dan robotik dalam pengeluaran pertanian pintar. Operasi automasi, pengendalian sistem robotik, dan penilaian keberkesanan penggunaan robotik dan jentera dalam pertanian pintar turut dibincangkan.

This course covers problem solving involving the use of automation and robotics systems in intelligent agricultural production. Automation operations, operation of robotic systems, and evaluation of the effectiveness of the use of robotics and machinery in intelligent agriculture are also discussed.

TKP4201	Pemindahan Teknologi Pertanian / <i>Transfer of Agricultural Technology</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi konsep, pembangunan, penilaian, perlindungan, serta isu dan cabaran semasa pemindahan teknologi pertanian.

This course encompasses concepts, development, evaluation, protection, current issues and challenges in agricultural technology transfer.

TKP4202	Komunikasi Pertanian / <i>Agricultural Communication</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi teori dan kaedah komunikasi pertanian, isu etika dan perundangan, strategi dan cabaran.

This course encompasses theories and methods in agricultural communication, legal and ethical issues, strategies and challenges.

TKP4203	Perencanaan Program Pengembangan Pertanian / <i>Programme Planning in Agricultural Extension</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini meliputi perencanaan pembentukan program pengembangan pertanian untuk meningkatkan produktiviti komuniti di kawasan terpilih. Pelaksanaan dan penilaian keberkesanan program oleh komuniti setempat juga diberi penekanan.

This course covers agricultural extension programme development planning to increase community productivity in selected areas. Implementation and assessment of programme effectiveness by the local community is also emphasized.

TKP4401	Analisis Dan Permodelan Spasial / <i>Spatial Modelling and Analysis</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penyelesaian masalah melibatkan pemodelan dan analisis spasial dalam pertanian pintar dengan menggunakan perisian Sistem Maklumat Geografi (GIS). Teknik model spasial, pemilihan analisis spasial dan teknik GIS yang sesuai berdasarkan saiz pengeluaran dan topografi ladang pengeluaran pertanian tertentu turut dibincangkan.

This course covers problem solving involving spatial modeling and analysis in smart agriculture using Geographic Information System (GIS) software. Spatial modeling techniques, selection of appropriate spatial analysis and GIS techniques based on production size and topography of specific agricultural production farms are also discussed.

TKP4402	Kejuruteraan Akuakultur / <i>Aquaculture Engineering</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi sistem akuakultur, pam pengudaraan, kolam dan tangki, sumber air, kualiti air, kadar alir, pengukuran aras dan rawatan air. Penekanan diberikan kepada unit penetasan, pemanenan, pemungutan dan pengangkutan tangkapan.

This course encompasses aquaculture systems, pump, aeration, ponds and tank, water supply, water quality, water flow, water level measurements and treatments. Emphasis is given to hatchery unit, harvesting, collection and transportation of catch.

TKP4403	Pembangunan Persekitaran Terkawal / <i>Development of Control Environment</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini memberi penekanan terhadap aspek perancangan, kawalan, penilaian prestasi dan pembuatan keputusan di dalam sesebuah syarikat perniagaantani dengan menggunakan maklumat perakaunan. Aspek teori dan praktikal dalam pengurusan kos perakaunan bagi memantau prestasi perniagaantani juga diberi penekanan.

This course emphasizes on planning, controlling, performance assessment and decision making in an agribusiness company by using accounting information. Theoretical and practical aspects in accounting cost management to monitor the performance of agribusiness are also emphasized.

TKP4404	Pengoperasian Sistem Sokongan Keputusan / <i>Decision Support System Operation</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi proses membuat keputusan, mengenalpasti sistem sokongan yang sesuai, dan aplikasinya dalam bidang pertanian pintar. Asas mengenalpasti faktor dalam aktiviti pengeluaran pertanian dan kesesuaian sistem untuk membuat keputusan turut dibincangkan.

This course covers the decision-making process and identify appropriate support systems, and their application in the field of smart agriculture. The basis of identifying factors in agricultural production activities and the suitability of the system for decision making are also discussed.

TKP4405	Pembangunan Aplikasi Mudah Alih / <i>Mobile Application Development</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi pembangunan aplikasi mudah alih dalam pertanian pintar. Pengadaptasian dan pengubahsuaian aplikasi mudah alih dalam pertanian secara aktiviti berkumpulan turut dibincangkan.

This course covers the development of mobile applications in smart agriculture. Adaptation and modification of mobile applications in agriculture in group activities are also discussed.

TKP4406	Pembangunan Pengairan Presis / <i>Development of Precision Irrigation</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penyelesaian masalah melibatkan pengairan dan penggunaan air yang presis. Teknik pengairan presis, pengendalian sistem pengairan, pemilihan model pengairan dan penderia yang sesuai berdasarkan saiz pengeluaran dan topografi ladang tanaman tertentu turut dibincangkan.

This course covers problem solving involving irrigation and precise water use. Precise irrigation techniques, operation of irrigation systems, selection of appropriate irrigation models and sensors based on production size and topography of specific crop fields are also discussed.

TKP4502	Pengoperasian Dron Lanjutan / <i>Advance Drone Operation</i>	3 (0+3)
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Prasyarat : TKP3502

Kursus ini merangkumi pengoperasian dron di lapangan mengikut jenis-jenis penderia. Pengurusan dan pengendalian penerbangan dron di lapangan dan penyelenggaraan dron dan mengadaptasi teknik penerbangan dron serta menunjuk kaedah pengendalian dron kepada komuniti teknologis juga turut dibincangkan.

This course covers the operation of drones in the field according to the types of sensors. Management and operation of drone flights in the field and maintenance of drones and adapting drone flight techniques as well as demonstrating drone handling methods to the technological community are also discussed.

TKP4801	Projek Tahun Akhir / <i>Final Year Project</i>	6 (0+6)
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Prasyarat : Tiada

Kursus ini merangkumi perancangan, pelaksanaan, analisis dan penterjemahan data, menyediakan dan membentangkan laporan projek tahun akhir.

This course covers the planning, implementation, analysis and translation of data, preparing and presenting final year project reports.

TKP4802	Kejuruteraan Akuakultur / <i>Aquaculture Engineering</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi sistem akuakultur, pam pengudaraan, kolam dan tangki, sumber air, kualiti air, kadar alir, pengukuran aras dan rawatan air. Penekanan diberikan kepada unit penetasan, pemanenan, pemungutan dan pengangkutan tangkapan.

This course encompasses aquaculture systems, pump, aeration, ponds and tank, water supply, water quality, water flow, water level measurements and treatments. Emphasis is given to hatchery unit, harvesting, collection and transportation of catch.

TKP4901	Pengurusan dan Pengoperasian Teknologi Pertanian Pintar / <i>Management and Operation of Smart Agricultural Technology</i>	12 (0+12)
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Prasyarat : Tiada

Kursus ini merangkumi perancangan, penilaian, pelaksanaan, merekabentuk, analisis dan penterjemahan data bagi projek pertanian pintar. Alternatif penyelesaian serta cadangan kos kepada permasalahan di industri turut dibincangkan.

This course covers the planning, evaluation, implementation, design, analysis and translation of data for smart farming projects. Alternative solutions as well as cost proposals to problems in the industry are also discussed.

TKP4902	Latihan Industri / <i>Industrial Attachment</i>	12 (0+12)
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Prasyarat : Tiada

Kursus ini melibatkan penempatan pelajar di ladang dan industri berkaitan pengurusan dan penggunaan teknologi sistem pertanian pintar di sektor kerajaan atau swasta. Pendekatan praktikal kepada keadaan pekerjaan dan amalan pengurusan sebenar dalam bidang teknologi pertanian pintar diberikan penekanan melalui projek jangka pendek.

This course involves the placement of students in farms and industries related to the management and use of intelligent agricultural system technology in the government or private sector. Practical approaches to employment conditions and actual management practices in the field of smart agricultural technology are given emphasis through short-term projects.